



WRRV301
2026

Functional Specification & Solution Blueprint

April 2026

Free HUB

Ventastik4_Devs

Mabulela Tumelo London

Mashau Lufuno Nathen

Mafunzwaini Umphe

Rametsi Tumelo Gift

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1 SYSTEM OVERVIEW

1.1 Problem Description

Many individuals with valuable abilities have difficulties in finding employment. Because their previous employers did not give them formal performance ratings or documented feedback, freelancers who have completed successful jobs in the past sometimes find themselves unable to demonstrate how reliable they are. These freelancers find it challenging to secure job opportunities as they are unable to establish a reliable reputation in the lack of visible verifiable reviews on their accounts.

Because they mainly use different social media platforms there is sometimes a communication gap between employers and freelancers. This communication typically leads to miscommunication, poor sourcing, fraud and unrecorded agreements.

The purpose of FreeHUB is to address the above issues by building a user-friendly system that will link freelancers with employers. Users of the FreeHUB system will be able to register and then set up a profile as a freelancer or employer. Freelancers will showcase their abilities, background, portfolio and hourly rates. Employers will advertise open positions. Freelancers will search for and select jobs that they can do. Employers will then interview freelancers and select the most qualified candidates for their project.

Throughout the project the system will enable internal messaging between employers and freelancers. To improve their profiles and reviews the employer and freelancers will rate one another when the job is done. Reviews and ratings are highly important since they help build a solid track record for potential employers or freelancers. Over time this reputation system will contribute to the development of a more robust freelancing economy where self-promotion becomes less important than confirmed performance and where both sides may interact with more trust and a lower chance of fraud or conflicts.

To ensure the system provides high-level data intelligence. It automatically calculates reputation scores by processing the reviews and ratings that both parties are required to submit after every project is completed. The system also includes a "Submit Proposal" feature, which allows freelancers to send formal bids for projects. An internal messaging system is provided to ensure all project communications are documented, while an administrator manages the platform and reviews dispute reports to maintain professional standards.

1.2 Business Rules / Assumptions

- A user can create a single account and choose to act as a freelancer or employer
- A user must have a unique email address
- A freelancer must complete their profile (skills, portfolio, experience and hourly rates)
- An employer must complete their profile with before posting any projects
- Employers must post projects with clear descriptions, timelines, and payment terms.
- A freelancer can submit one proposal per project.
- A freelancer cannot submit a proposal to their own project
- An employer can select one or more freelancer per project
- Once a freelancer is selected for a project, the project status changes from "Open" to "Progress"

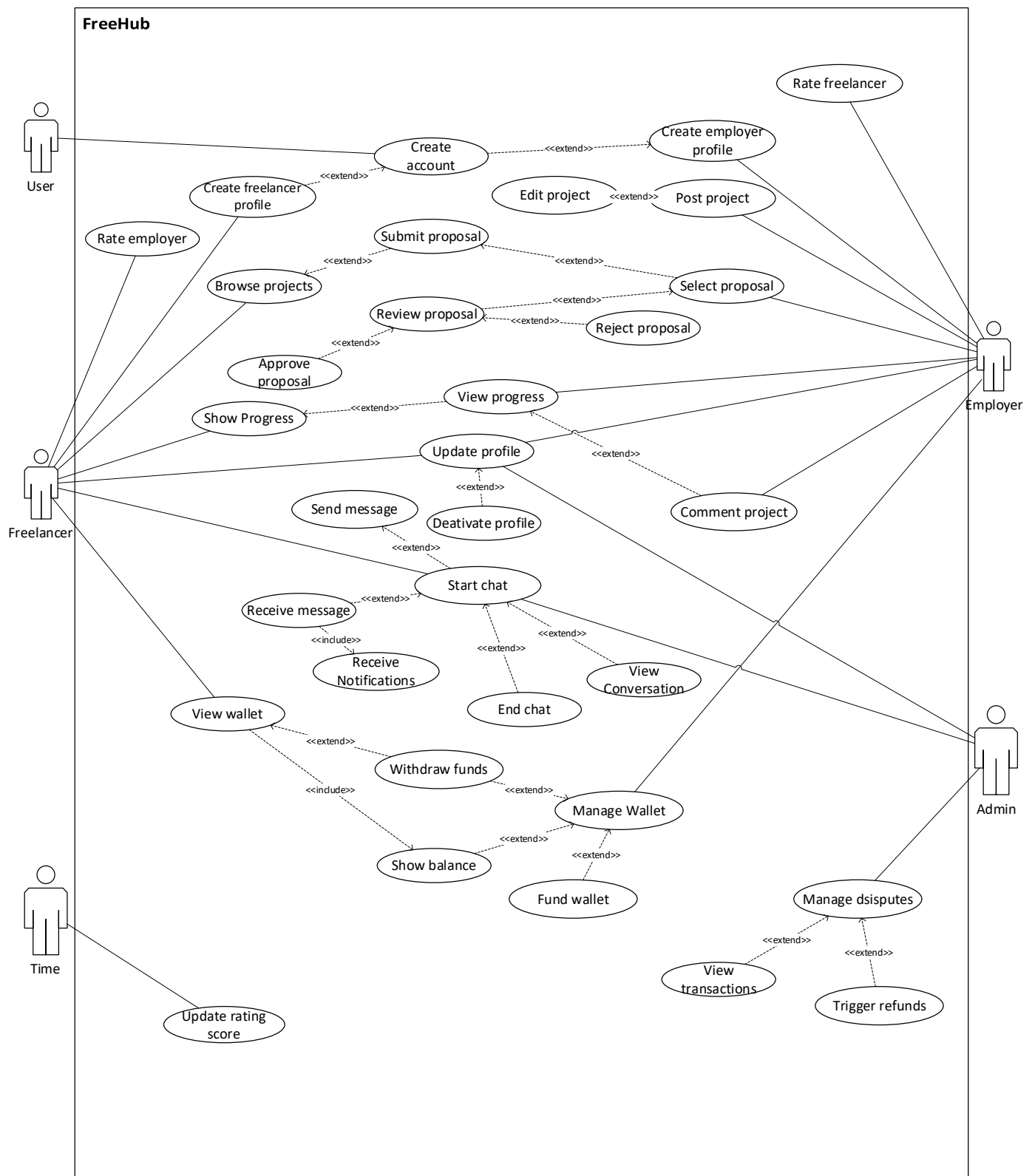
- Both the freelancer and employer must submit a rating after a project is completed to maintain trust and reputation (e.g., One-to-five stars)
- Ratings may be accompanied by optional written evaluations.
- Admin profiles are created manually in the database
- Admins oversee dispute resolution, monitor fraudulent activity, and ensure compliance with platform policies.
- Communication between users must occur through the internal messaging system to ensure documentation.
- Assume that users have reliable internet access and basic digital literacy.
- Assume that payment handling will be internal and processed directly by the system in this phase.
- The "Submit Proposal" function is assumed to be the primary method for a freelancer to provide an employer with project timelines and cost estimates.

1.3 System Constraints

- The system is web-based only.
- Continuous internet access is mandatory for all users. Offline functionality, caching, or synchronization features are not supported in this program.
- Only administrators have access to backend system management features.
- Messaging is limited to text-based communication (no file/image sharing).
- Messaging is limited to system users only.
- Ratings are restricted to stars between One and Five stars.
- User passwords must be stored securely using hashing techniques.
- Users cannot delete their accounts while involved in an active project.
- File uploads (e.g., Portfolio images) are limited by size and type.
- The system relies on external hosting services and SQL Server databases availability.
- The system must comply with POPIA and basic data protection standards.
- Advanced security mechanisms.
- Employers and freelancers are the only sources of project and user data.
- Videos and audio files are not permitted in freelancer profile, but text descriptions of skills and experience are permitted.
- The system will be optimized for modern browsers (Chrome, Firefox, Edge) but may not fully support outdated versions or specialized hardware configurations.

2 FUNCTIONAL REQUIREMENTS

2.1 Initial Use Case Model



Final Use Case Model

[Replace this text with the Use Case diagram – this must be drawn in Visio.]

2.2 Use Case Glossary and Responsibilities

Allocated to Crew Member: Umphe Mafunzwaini	
Use Case Id	Use Case Name
A100	Create Account
A200	Create Freelancer Profile
A300	Create Employer Profile
A400	Deactivate Profile
A500	Update Profile
A600	Deactivate Account
A700	View Progress
A800	Show Progress

Allocated to Crew Member: Tumelo London Mabulela	
Use Case Id	Use Case Name
B100	Browse Project
B200	Post Project
B300	Edit Project
B400	Comment Project
B500	Submit Proposal
B600	Select Proposal
B700	Review Proposal
B800	Approve Proposal
B900	Reject Proposal

Allocated to Crew Member: Lufuno Naithern Mashau	
Use Case Id	Use Case Name
C100	Start Chat
C200	Send Message
C300	Receive Message
C400	Receive Notification
C500	End Chat
C600	View Conversation
C700	Rate Employer
C800	Rate Freelancer
C900	Update Rating Score

Allocated to Crew Member: Tumelo Gift Rametsi

Use Case Id	Use Case Name
D100	Manage Wallet
D200	View Wallet
D300	Show Balance
D400	Withdraw Funds
D500	Fund Wallet
D600	View Transactions
D700	Manage Disputes
D800	Trigger Refunds

2.3 Use Case Narratives & UI Designs

2.3.1 Team Designs

2.3.1.1 Team User Interface Guidelines

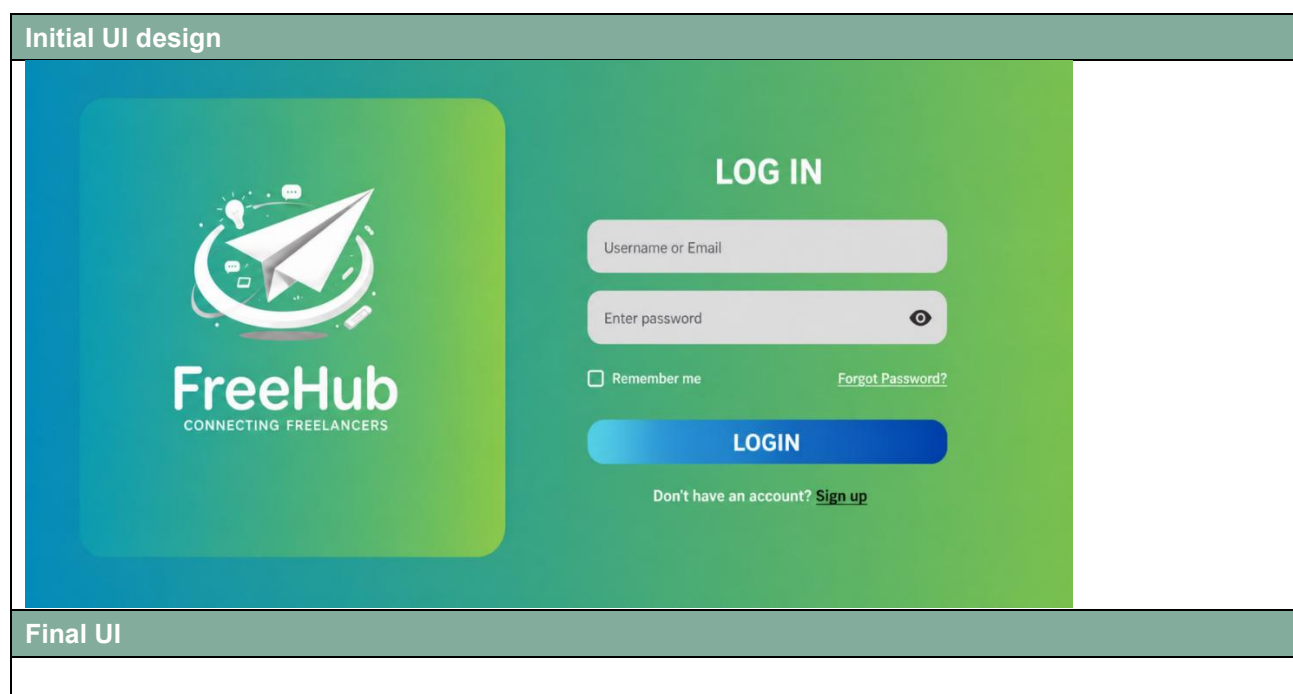
The User Interface must be kept simple, clean, and modern. This will help attract young users while remaining accessible to individuals with varying levels of experience using mobile applications.

The colour scheme of the UI is themed around nature, incorporating muted and warm tones derived from the selected palette, including deep green, muted green, soft neutral, and light beige.



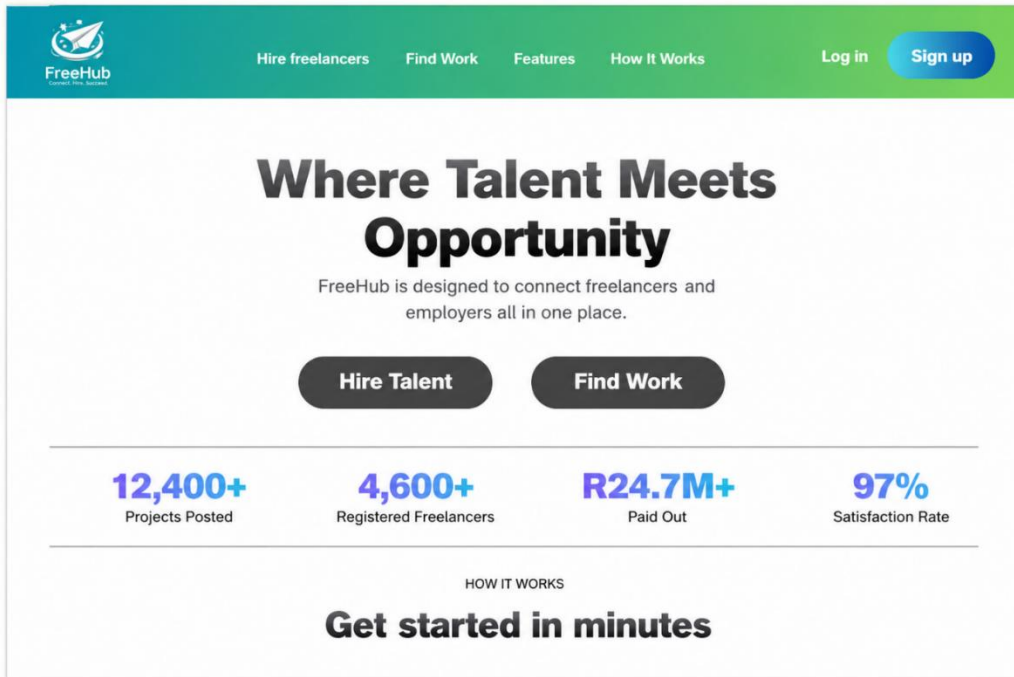
The fonts used were Arial. The fonts were chosen for their legibility in varying sizes and colours. To align with a modern and approachable style, shapes used in the UI have rounded edges wherever possible.

2.3.1.2 Login / Logout Feature



2.3.1.3 Dashboard / Landing Page / Main Page

Initial UI design



Final UI

2.3.2 Individual Crew Member Use Case Narratives & UI Designs

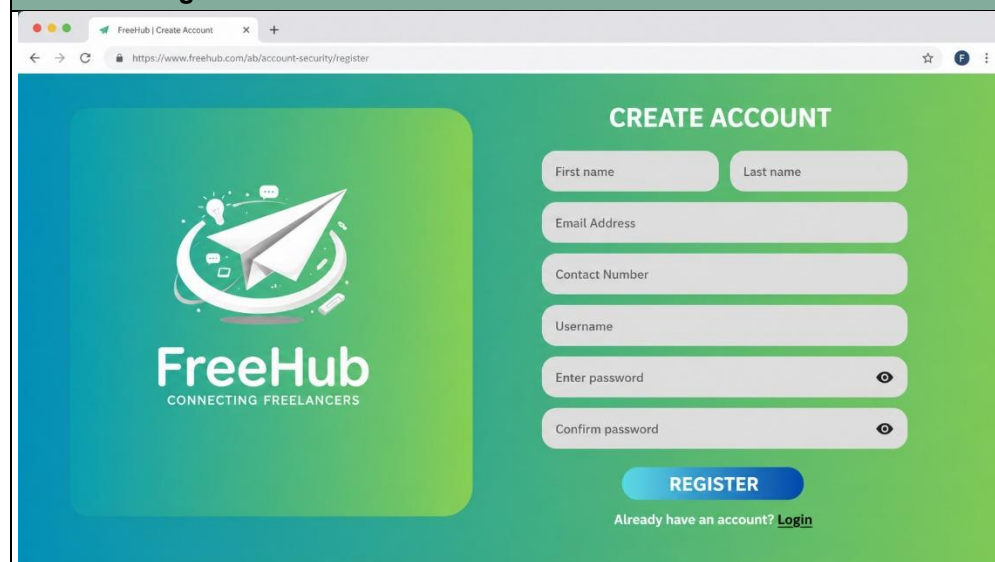
2.3.2.1 Designed by Umphe Mafunzwaini

Use Case ID	Use Case Name
A100	Create Account
Actors	
User	
Pre-Conditions	You can access Free HUB via a browser.
Triggers	User will select a "Sign Up" on the login page.
Post-Conditions	A new user account is created.

Basic Flow of Events / Main Success Scenario

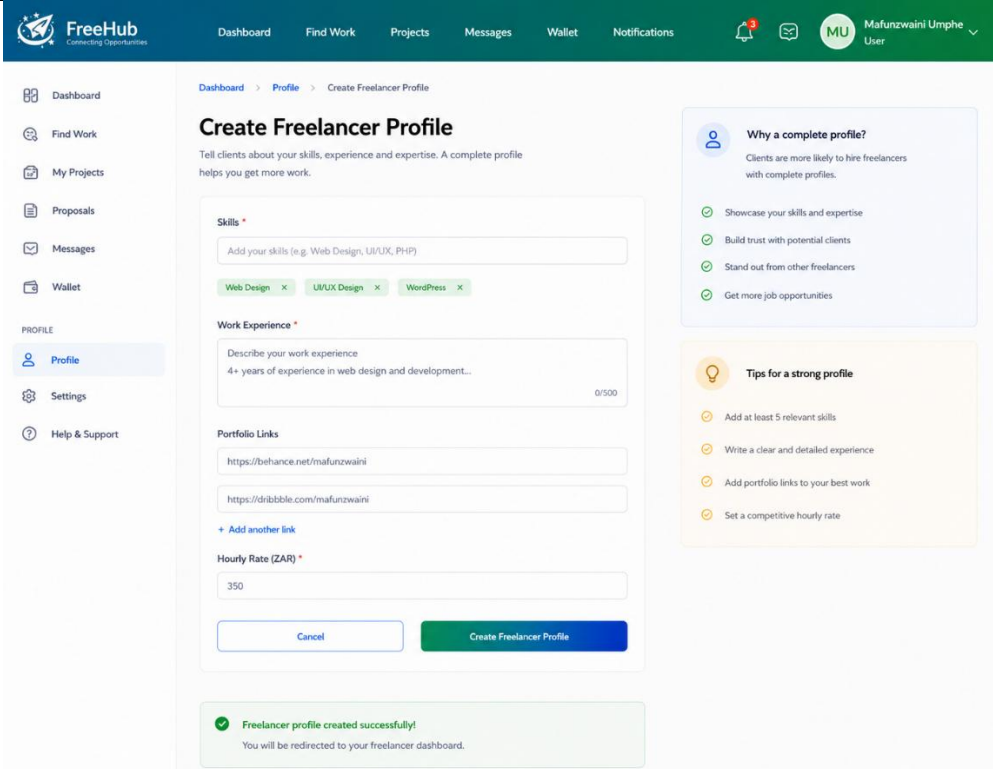
2. The User selects the Sign-Up option on the login page.
3. The system displays a registration form with fields for first name, last name, email, contact, username and password
4. The User enters the required details into the form fields.
5. The system validates that all required fields are complete and the input is valid (format of email, unique username and password meets requirements).
6. The User selects the Create Account option.
7. The system creates the user's account.
8. The system displays a message confirming successful account creation.
9. The system redirects the user to the login interface.

Initial UI Design



Final Implemented UI Design

Use Case ID	Use Case Name
A200	Create Freelancer Profile
Actors	

User	
Pre-Conditions	The user has a registered account and is logged in.
Triggers	The user selects “Become a Freelancer” from the dashboard.
Post-Conditions	A freelancer profile is linked to the user’s account.
Basic Flow of Events / Main Success Scenario	
1. The user selects “Become a Freelancer” from the dashboard. 2. The system displays a form with fields for skills, work experience, portfolio links, and hourly rate. 3. The user enters the required details into the form fields provided. 4. The system validates that all required fields are complete and input is valid (e.g. at least one skill). 5. The user selects the “Create Freelancer Profile” option. 6. The system creates a freelancer profile and links it to the user’s account. 7. The system displays a message confirming successful profile creation. 8. The system redirects the freelancer to the freelancer dashboard.	
Initial UI Design	
	
Final Implemented UI Design	

Use Case ID	Use Case Name
A300	Create Employer Profile
Actors	
User	

Pre-Conditions	The User has a registered account and is logged in.
Triggers	The user selects “Become an Employer” from the dashboard.
Post-Conditions	An employer profile is linked to the user’s account.

Basic Flow of Events / Main Success Scenario

1. The User selects “Become an Employer” from the dashboard.
2. The system displays a form with fields for company name, industry, company description and contact email.
3. The User enters the required information into the form fields.
4. The system validates that all required fields are complete and the input is valid.
5. The user selects the “create Employer Profile” option.
6. The system creates an employer profile and links it to the user’s account.
7. The system displays a message confirming successful profile creation.
8. The system redirects the actor to the employer dashboard.

Initial UI Design

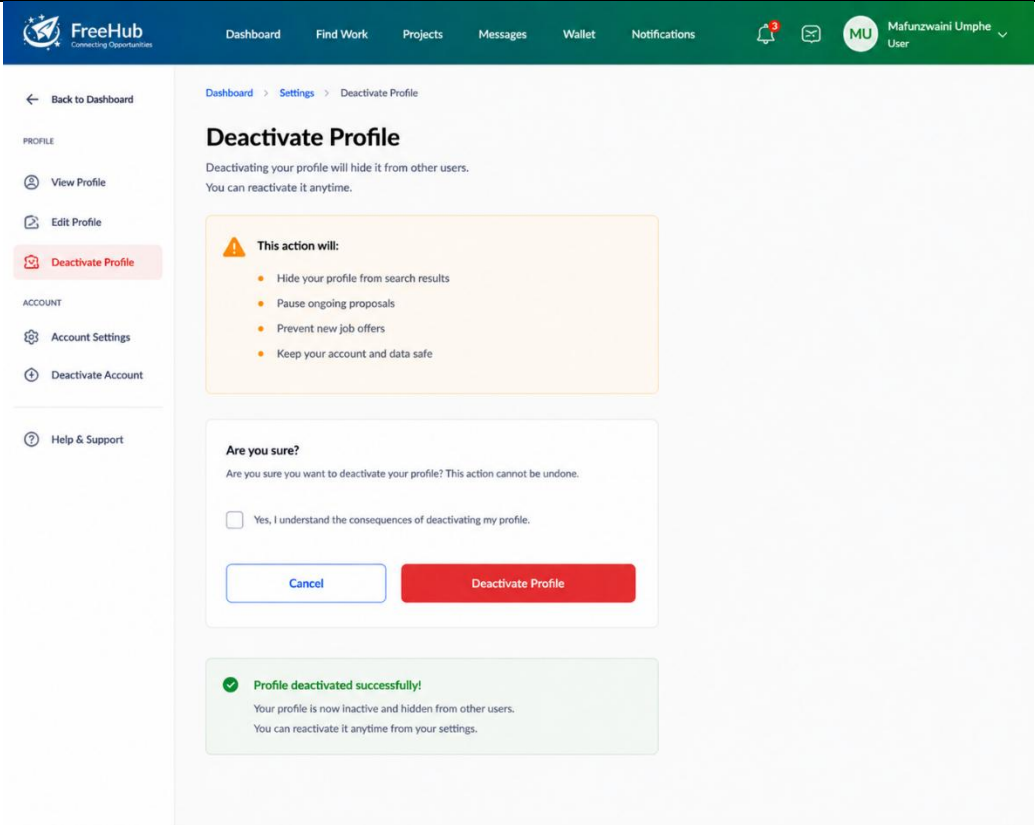
The screenshot shows the 'Create Employer Profile' page on the FreeHub platform. The top navigation bar includes links to Dashboard, Find Work, Projects, Messages, Wallet, and Notifications. The left sidebar lists various user options. The main form area contains the following sections:

- Company Name ***: A text input field with 'Umphe Solutions (Pty) Ltd' entered.
- Industry ***: A dropdown menu with 'Information Technology' selected.
- Company Description ***: A text area with the description 'We provide innovative digital solutions and IT services to help businesses grow.'
- Contact Email ***: A text input field with 'hello@umphe-solutions.co.za' entered.

At the bottom of the form are 'Cancel' and 'Create Employer Profile' buttons. A green success message at the bottom states: 'Employer profile created successfully! You will be redirected to your employer dashboard.'

Final Implemented UI Design

Use Case ID	Use Case Name
A400	Deactivate Profile
Actors	
User	
Pre-Conditions	The user is logged in and has an active profile.
Triggers	The user selects “Deactivate Profile”.

Post-Conditions	The profile becomes inactive.
Basic Flow of Events / Main Success Scenario	
1. The user selects Deactivate Profile. 2. The system displays a confirmation message. 3. The user confirms deactivation. 4. The system deactivates the profile. 5. The system displays a successful deactivation message.	
Initial UI Design	
	
Final Implemented UI Design	

Use Case ID	Use Case Name
A500	Update Profile
Actors	
User	
Pre-Conditions	The user is logged in and has an existing profile.
Triggers	The user selects "Update Profile".
Post-Conditions	The profile details are updated.
Basic Flow of Events / Main Success Scenario	
1. The user selects Update Profile. 2. The system displays the current profile details.	

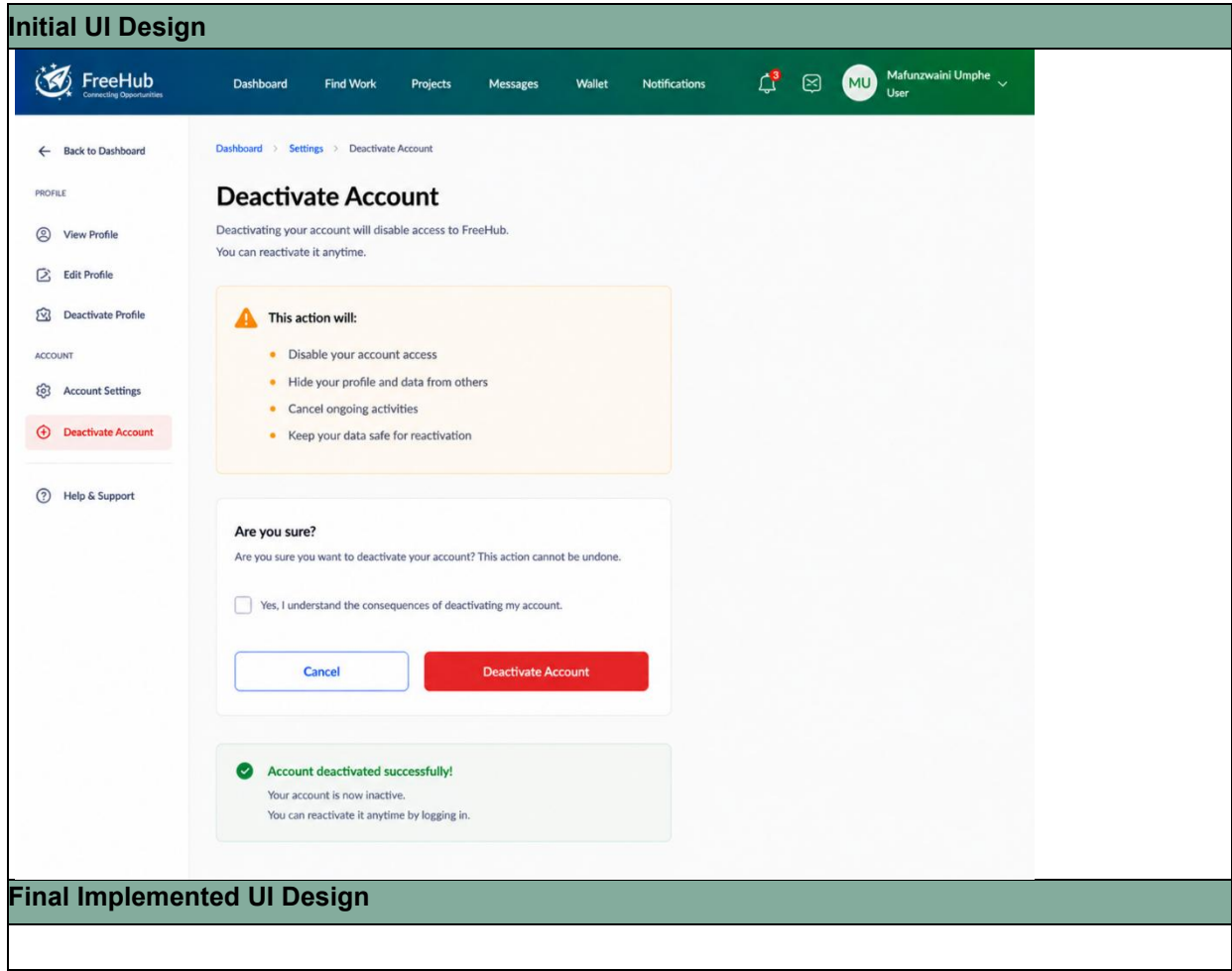
3. The user edits the required information.
4. The system validates the updated details.
5. The user selects Save Changes.
6. The system updates the profile.
7. The system displays a confirmation message.

Initial UI Design

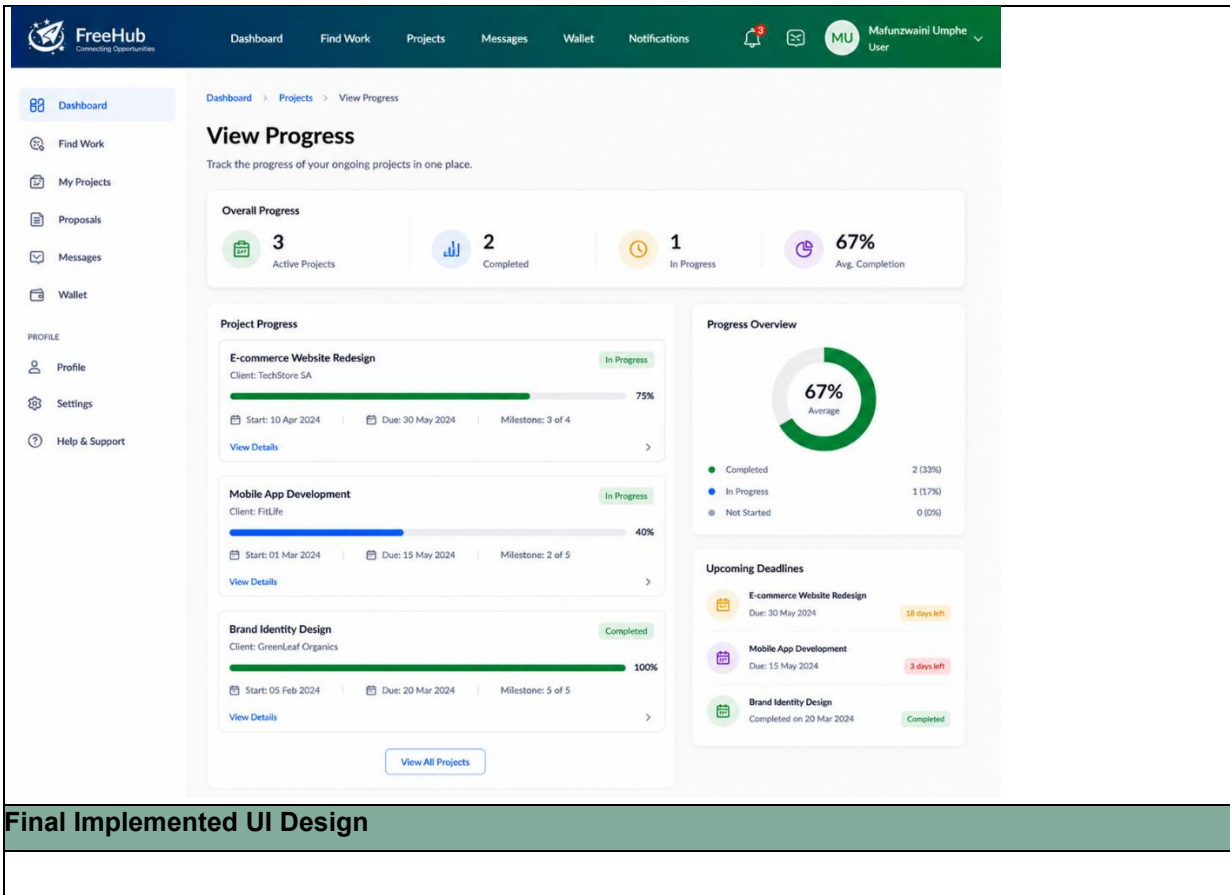
The screenshot shows the 'Update Profile' form in the FreeHub application. The form is titled 'Update Profile' and includes a sub-header 'Update your information and keep your profile up to date.' The form is divided into two main sections: 'PROFILE' and 'ACCOUNT'. The 'PROFILE' section contains fields for 'Full Name', 'Headline', 'Bio', 'Skills', and 'Hourly Rate (ZAR)'. The 'ACCOUNT' section contains 'Account Settings' and 'Deactivate Account'. The 'Full Name' field is filled with 'Mafunzwaini Umphe'. The 'Headline' field is filled with 'Web Designer & Frontend Developer'. The 'Bio' field is filled with 'Passionate web designer and frontend developer with a focus on creating clean, user-friendly digital experiences.' The 'Skills' field is filled with 'Web Design', 'HTML', 'CSS', and 'JavaScript'. The 'Hourly Rate (ZAR)' field is filled with '400'. The 'Profile Picture' section shows a preview of the profile picture and a 'Change Photo' button. The 'Save Changes' button is highlighted in blue. A confirmation message at the bottom states 'Profile updated successfully! Your changes have been saved.'

Final Implemented UI Design

Use Case ID	Use Case Name
A600	Deactivate Account
Actors	
User	
Pre-Conditions	The user is logged in and has an active account.
Triggers	The user selects "Deactivate Account".
Post-Conditions	The account becomes inactive.
Basic Flow of Events / Main Success Scenario	
1. The user selects Deactivate Account. 2. The system displays a confirmation message. 3. The user confirms deactivation. 4. The system deactivates the account. 5. The system displays a successful deactivation message.	

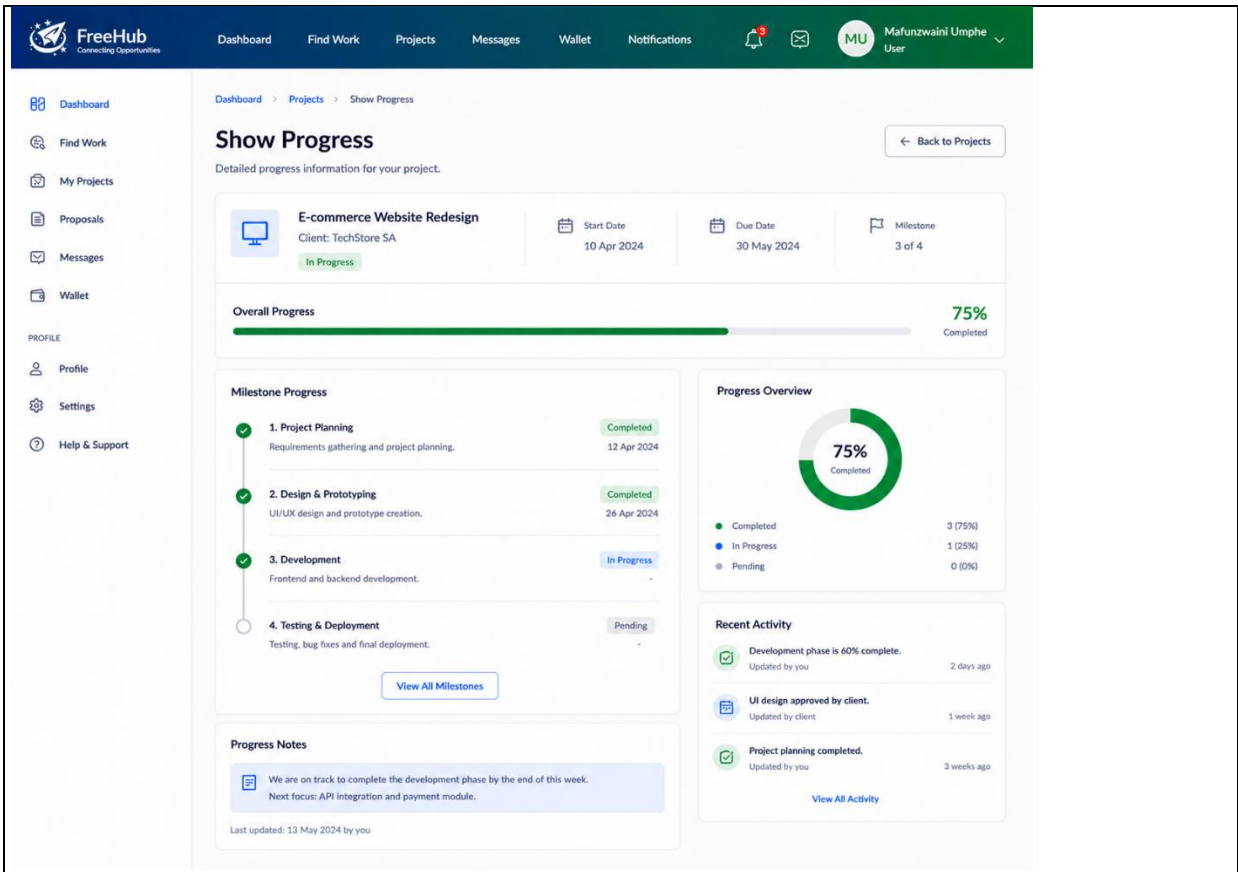


Use Case ID	Use Case Name
A700	View Progress
Actors	
User	
Pre-Conditions	The user is logged into the system.
Triggers	The user selects “View Progress”.
Post-Conditions	The user’s project progress is displayed.
Basic Flow of Events / Main Success Scenario	
1. The user selects View Progress. 2. The system retrieves progress information. 3. The system displays the progress details.	
Initial UI Design	



Final Implemented UI Design

Use Case ID	Use Case Name
A800	Show Progress
Actors	
System	
Pre-Conditions	Progress information exists.
Triggers	The user requests to view progress.
Post-Conditions	Progress is shown to the user.
Basic Flow of Events / Main Success Scenario	
- The system retrieves the latest progress status. - The system calculates the current progress percentage. - The system displays progress to the user.	
Initial UI Design	



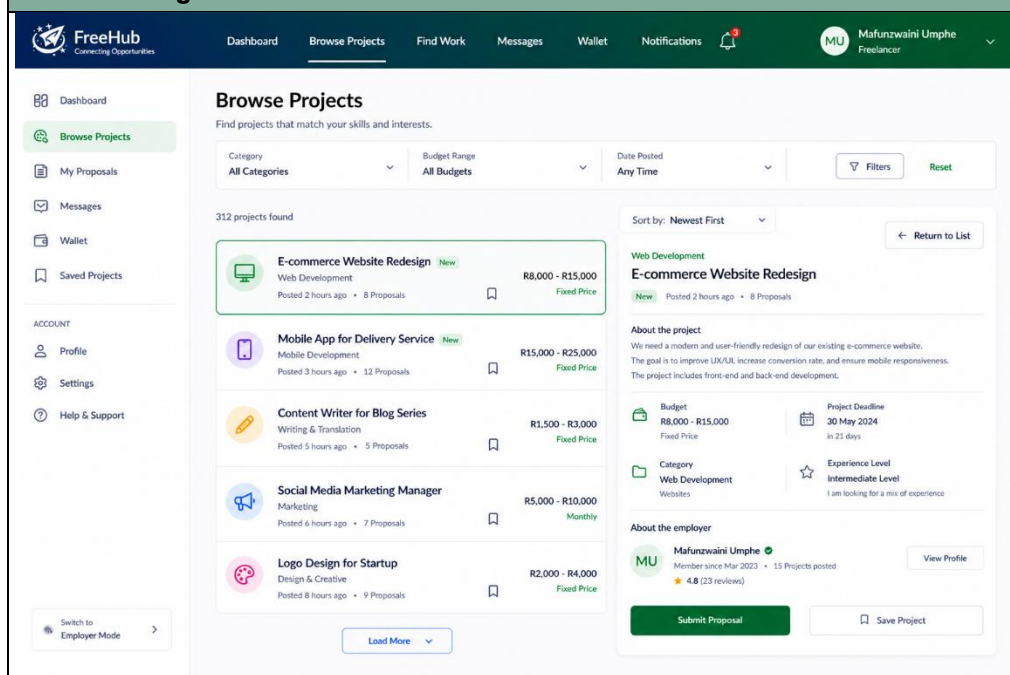
Final Implemented UI Design

2.3.2.2 Designed by Tumelo London Mabulela

Use Case ID	Use Case Name
B100	Browse Projects
Actors	
Freelancer	
Pre-Conditions	The freelancer is logged in and has a complete freelancer profile.
Triggers	The freelancer selects “Browse Projects” from the navigation menu.
Post-Conditions	The system displays all available projects list.
Basic Flow of Events / Main Success Scenario	

1. The freelancer selects “Browse Projects” from the navigation menu.
2. The system displays a list of open projects with filter options (category, budget range, date posted).
3. The freelancer applies filters or scrolls through the project list.
4. The system updates the displayed project list based on the selected filters.
5. The freelancer selects a project to view full details about the project.
6. The system displays a full project description, budget deadline and employer information.
7. The freelancer selects the Return-to-list option.
8. The system returns to the project list displayed in step 2.
9. The freelancer selects “load More” to view additional projects.
10. The system loads and displays the next page of projects.

Initial UI Design



Final Implemented UI Design

Use Case ID	Use Case Name
B200	Post Project
Actors	
Employer	
Pre-Conditions	The employer is logged in and has an employer profile.
Triggers	The employer selects “Post Project”.
Post-Conditions	A new project is created and displayed to freelancers.
Basic Flow of Events / Main Success Scenario	
1. The employer selects Post Project. 2. The system displays a project form.	

3. The employer enters project title, description, category, budget, and deadline.
4. The system validates the project details.
5. The employer selects Submit.
6. The system creates the project.
7. The system displays a confirmation message.

Initial UI Design

FreeHub Connecting Opportunities

Dashboard Find Work Projects Messages Wallet Notifications

ML Mabulela London Employer

Dashboard > My Projects > Post Project

Post Project

Fill in the details below to create a new project and share it with freelancers.

1 Project Title: E-commerce Website Redesign

2 Category: Web Development

3 Project Description: I need a modern, responsive redesign of my existing e-commerce website. The new design should improve user experience, site speed, and mobile responsiveness. It should include product pages, shopping cart, secure checkout, and integration with payment gateways.

4 Budget Type: Fixed Price

5 Budget Amount (ZAR): R 25,000

6 Project Deadline: 30 Jun 2024

7 Experience Level: Intermediate

8 Skills & Expertise: HTML, CSS, JavaScript, Shopify, UI/UX Design

9 Attachments (Optional): Drag & drop files here or click to upload

Cancel Preview Project Submit Project

Project posted successfully! Your project is now visible to freelancers.

Tips for a Great Project Post

- Be clear and specific:** A detailed description helps freelancers understand your needs.
- Set a realistic budget:** Fair budgets attract better quality freelancers.
- Choose the right category:** Selecting the right category gets your project in front of the right experts.
- Add key skills:** List important skills to match with qualified freelancers.
- Set a reasonable deadline:** Give enough time for quality work without unnecessary delays.
- Secure & Safe:** Your project is protected by our escrow system. Release payment only when you're satisfied.

Final Implemented UI Design

Use Case ID	Use Case Name
B300	Edit Project
Actors	
Employer	
Pre-Conditions	The employer is logged in and has already posted a project.
Triggers	The employer selects "Edit Project".
Post-Conditions	The project details are updated.
Basic Flow of Events / Main Success Scenario	
1. The employer selects one of their posted projects. 2. The employer selects Edit Project. 3. The system displays the project details form. 4. The employer updates the necessary information. 5. The system validates the updated details. 6. The employer saves the changes. 7. The system updates the project. 8. The system displays a confirmation message.	
Initial UI Design	



FreeHub

Connecting Opportunities

Dashboard

Browse Projects

Find Work

Messages

Wallet

Notifications



ML

Mabulela London

Employer



Dashboard

My Projects

Post Project

MANAGE PROJECTS

My Projects

Drafts

Archived

ACCOUNT

Profile

Settings

Help & Support

Switch to Freelancer Mode



Dashboard > My Projects > Edit Project

Edit Project

Update your project details below and save the changes.

Project Title *

E-commerce Website Redesign

Category *

Web Development

Project Description *

B

I

U

We need a modern and user-friendly redesign of our existing e-commerce website.
The goal is to improve UX/UI, increase conversion rate, and ensure mobile responsiveness.
The project includes front-end and back-end development.

196/2000

Budget Type *

Fixed Price

Hourly

Budget Amount *

R

12,000

Project Deadline *

30 May 2024

Experience Level *

Intermediate Level

in 21 days

Skills & Expertise (Optional)

HTML

CSS

JavaScript

PHP

MySQL

Cancel

Save Changes

Tips for Editing Projects

Be Clear & Specific

Update your project title and description to attract the right talent.

Set a Realistic Budget

Ensure your budget reflects the scope of the work.

Adjust Deadline

Give enough time for quality work to be delivered.

Add Relevant Skills

List the key skills to get proposals from qualified freelancers.

Review Before Saving

Double-check all details before updating your project.

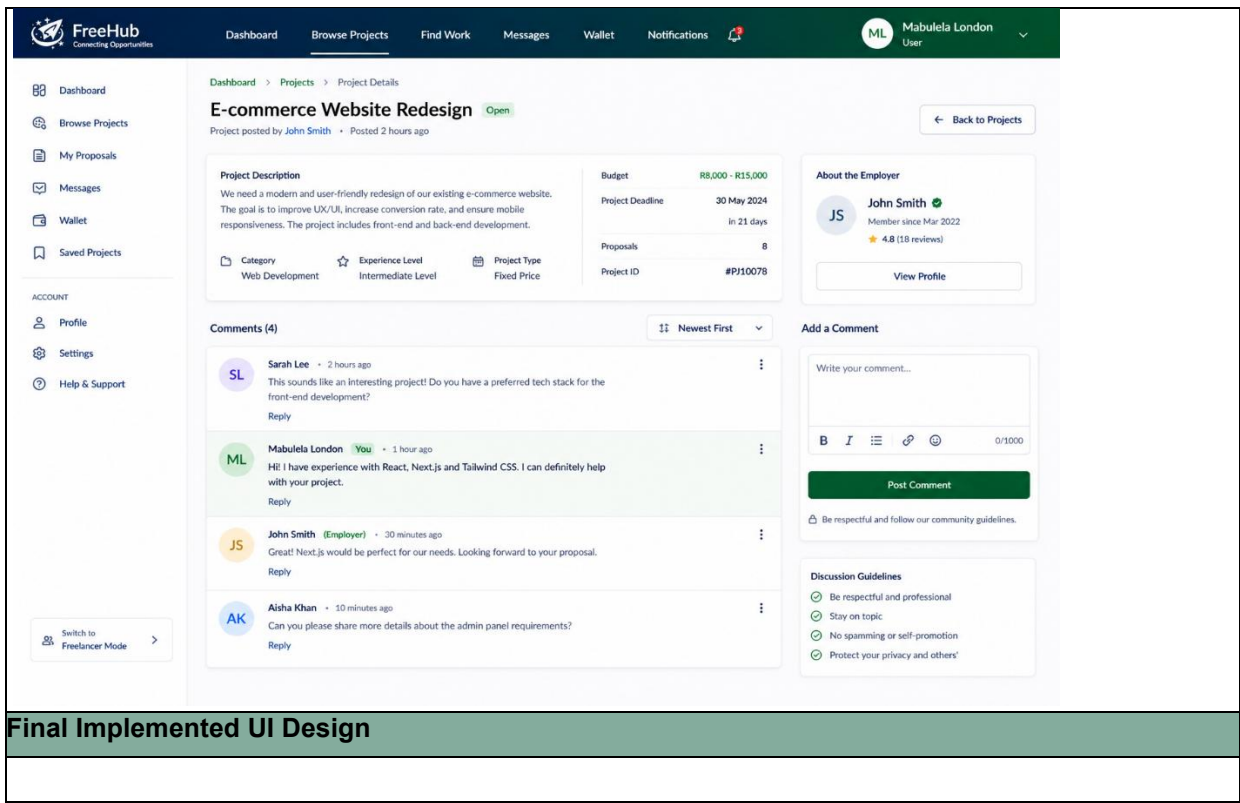
Project updated successfully!

Your changes have been saved.

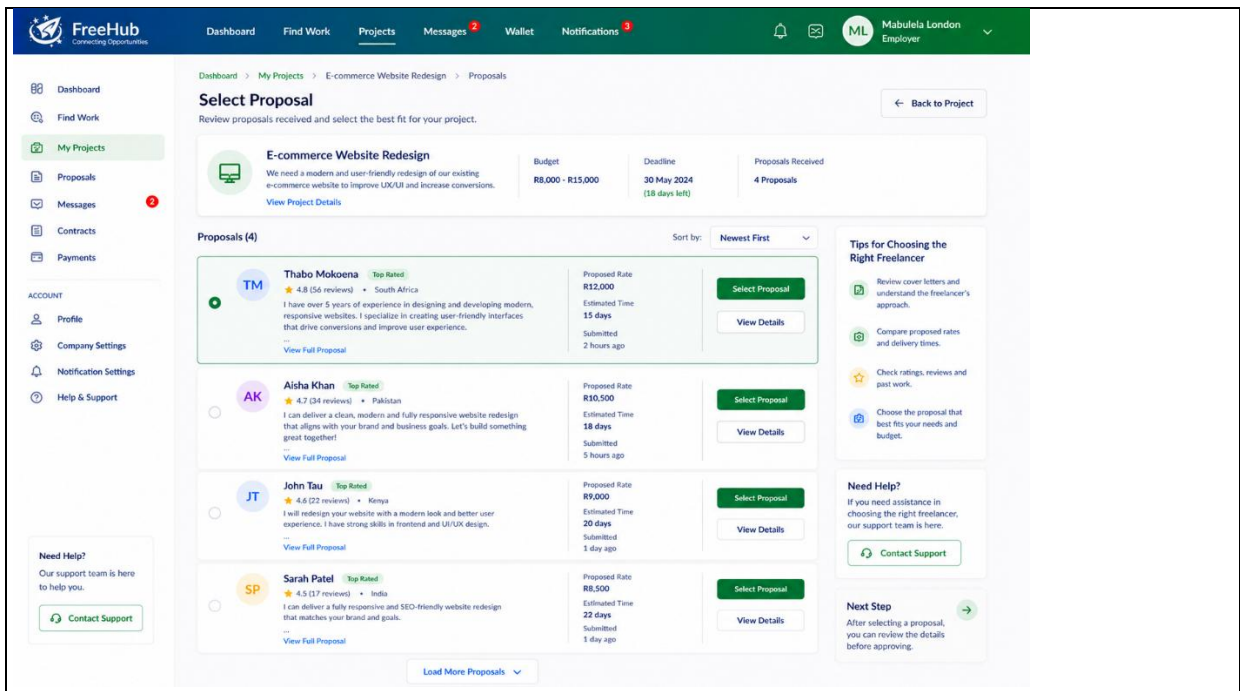


Final Implemented UI Design

Use Case ID	Use Case Name
B400	Comment Project
Actors	
User	
Pre-Conditions	The user is logged in and has access to a project.
Triggers	The user selects “Comment”.
Post-Conditions	The comment is added to the project.
Basic Flow of Events / Main Success Scenario	
1. The user opens a project. 2. The user selects Comment. 3. The system displays a comment field. 4. The user enters a comment. 5. The user submits the comment. 6. The system saves the comment. 7. The system displays the comment under the project.	
Initial UI Design	

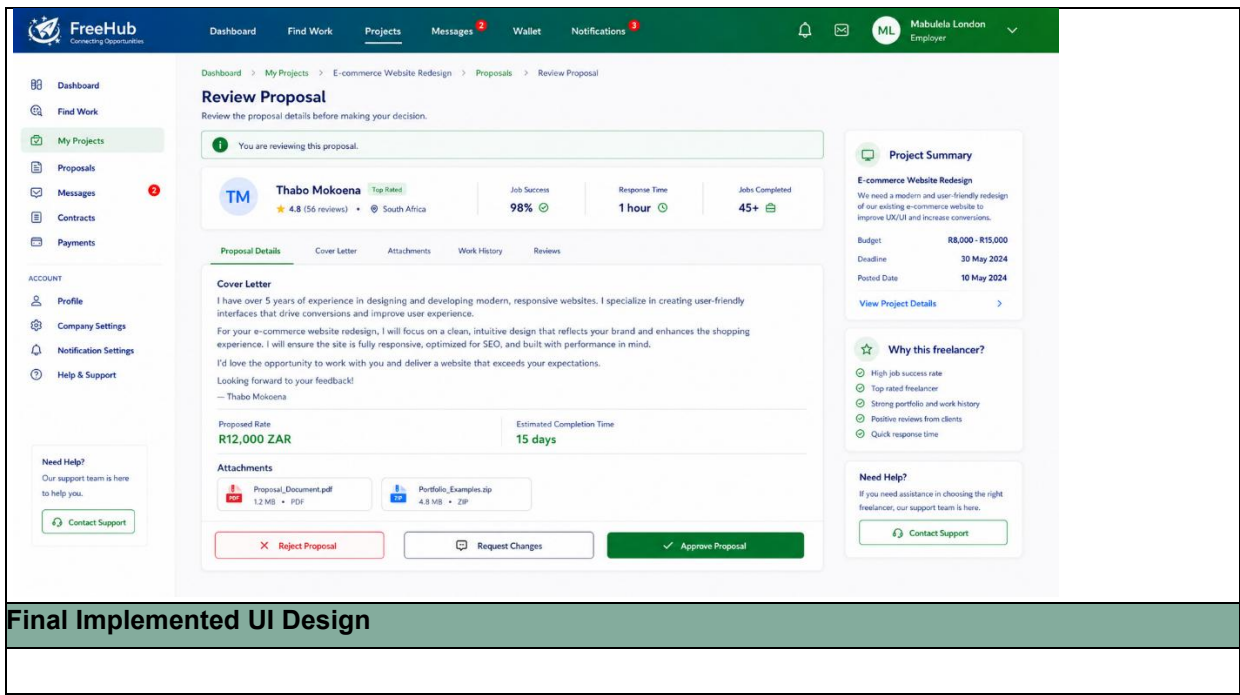


Use Case ID	Use Case Name
B500	Submit Proposal
Actors	
Freelancer	
Pre-Conditions	The freelancer is logged in the system and has viewed a project.
Triggers	The freelancer selects the “Submit Proposal” on a project details page.
Post-Conditions	A new proposal is created and linked to the project and freelancer.
Basic Flow of Events / Main Success Scenario	
1. The freelancer selects Submit Proposal on the project details page. 2. The system displays a proposal form with fields for cover letters, proposed rate, and estimated completion time. 3. The freelancer enters the required information into the form fields. 4. The system validates that all required fields are complete and the input is valid. 5. The freelancer selects the “Submit Proposal” option. 6. The system creates the proposal and links it to the project and freelancer. 7. The system displays a message confirming successful proposal submission. 8. The system returns the freelancer to the project details page.	
Initial UI Design	

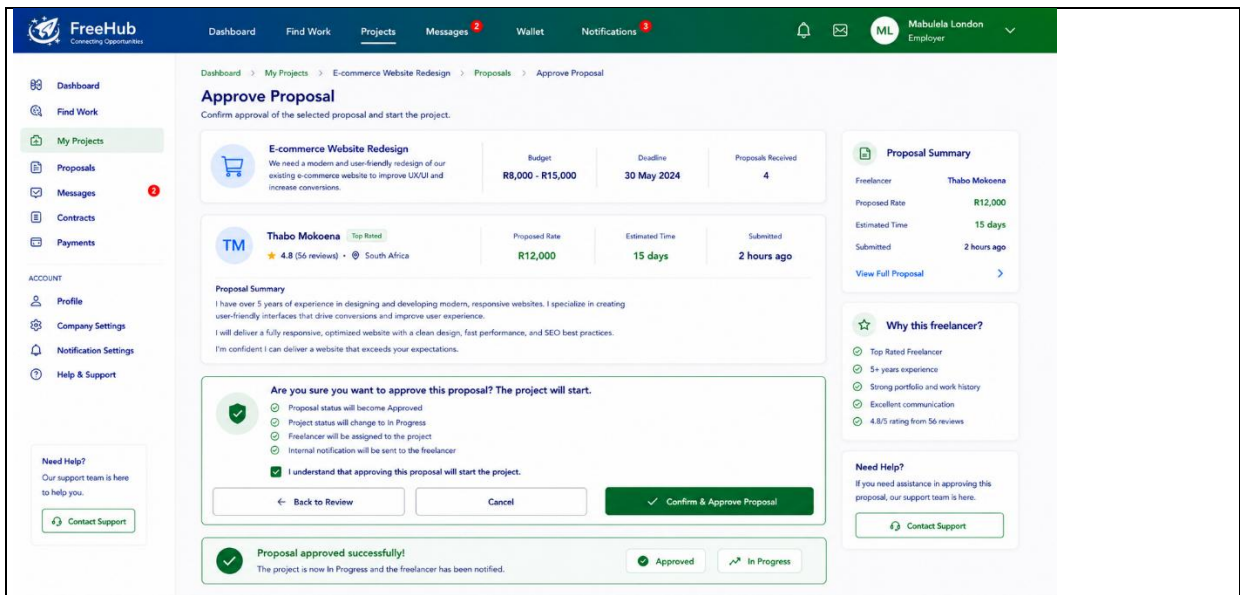


Final Implemented UI Design

Use Case ID	Use Case Name
B600	Select Proposal
Actors	
Employer	
Pre-Conditions	The employer is logged in and has a project with at least one submitted proposal.
Triggers	The employer opens the project's proposal list.
Post-Conditions	The selected proposal is marked for reviewing.
Basic Flow of Events / Main Success Scenario	
1. The employer opens the proposal list for a specific project. 2. The system displays a list of proposals showing freelancer names, proposed rates, and cover letter. 3. The employer browses through the proposal list. 4. The employer selects a proposal by clicking on it. 5. The system highlights the selected proposal. 6. The employer confirms the selection to proceed to full review. 7. The system proceeds with the proposal review interface.	
Initial UI Design	

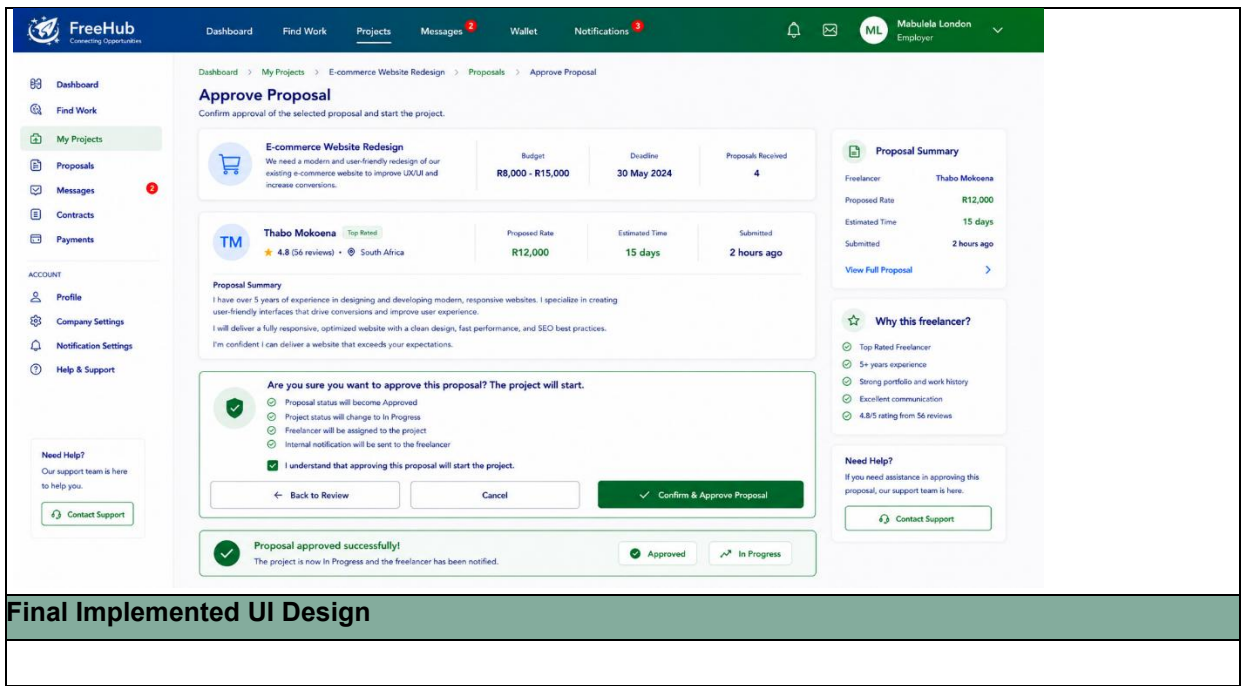


Use Case ID	Use Case Name
B700	Review Proposal
Actors	
Employer	
Pre-Conditions	A proposal has been selected by the employer.
Triggers	The employer selects “Review Proposal” from the selected proposal view.
Post-Conditions	The employer can approve, reject or request changes to the proposal.
Basic Flow of Events / Main Success Scenario	
1. The employer selects “Review Proposal” from the selected proposal view. 2. The system displays the full proposal details including complete cover letter, proposed rate, estimated timeline and a link to the freelancer's profile. 3. The employer reviews all proposal information carefully. 4. The employer selects one of the following actions which is Approve, Reject or Request changes. 5. The system processes the selected action. 6. The system displays a confirmation message for the action taken. 7. The system redirects the employer to the proposal list.	
Initial UI Design	



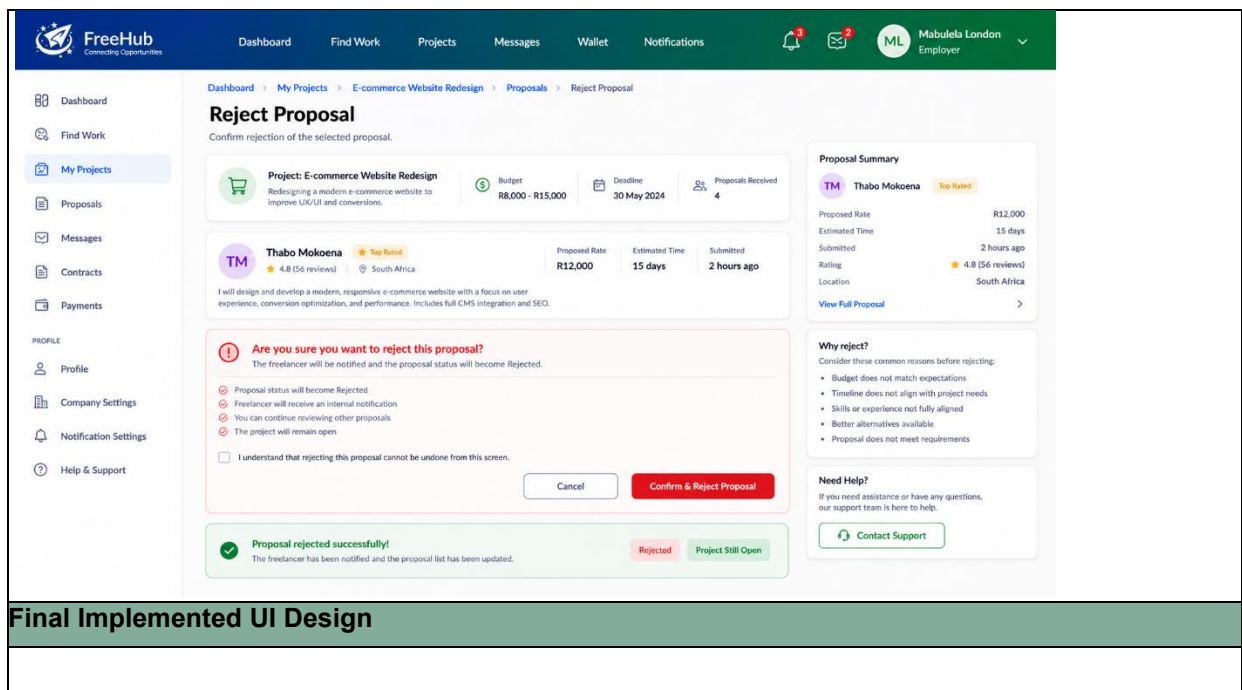
Final Implemented UI Design

Use Case ID	Use Case Name
B800	Approve Proposal
Actors	
Employer	
Pre-Conditions	The Proposal is currently under review.
Triggers	The employer selects Approve while reviewing the proposal
Post-Conditions	The status of the proposal becomes Approved, the project status changes to in progress and the freelancer is assigned.
Basic Flow of Events / Main Success Scenario	
1. The employer selects the Approve option while reviewing the proposal. 2. The system displays a confirmation dialog asking “are you sure you want to approve this proposal? The project will start” 3. The employer confirms the approval. 4. The system updates the proposal status to “Approved”. 5. The system updates the project status to “In progress”. 6. The system assigns the freelancer to the project. 7. The system sends an internal message notification to the freelancer. 8. The system displays a success pop up message. 9. The system redirects the employer to the project management page.	
Initial UI Design	



Final Implemented UI Design

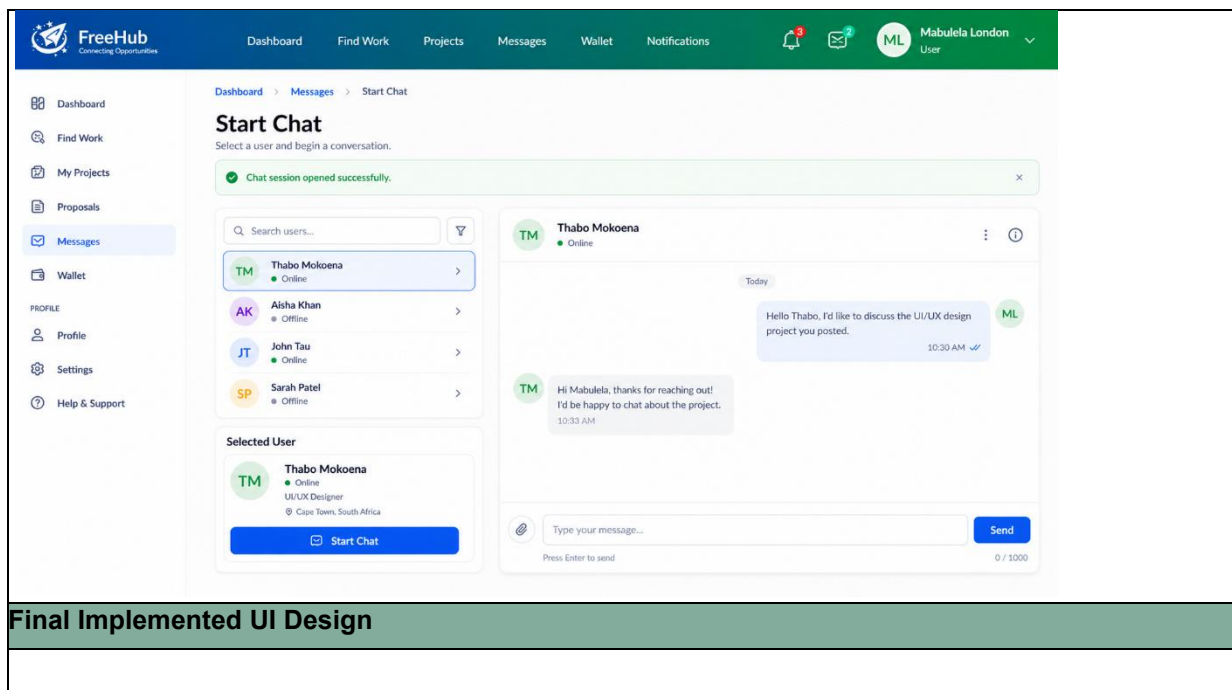
Use Case ID	Use Case Name
B900	Reject Proposal
Actors	
Employer	
Pre-Conditions	The proposal is currently being reviewed.
Triggers	The employer selects “reject’ while reviewing the proposal.
Post-Conditions	The proposal status becomes “rejected” and the freelancer is notified.
Basic Flow of Events / Main Success Scenario	
1. The employer selects the rejection while reviewing a proposal. 2. The system displays a confirmation message asking, “Are you sure you want to reject this proposal?”. 3. The employer confirms the rejection. 4. The system updates the proposal status to “rejected”. 5. The system sends an internal message notification to the freelancer. 6. The system displays a confirmation pop up message. 7. The system returns the actor to the proposal list.	
Initial UI Design	



Final Implemented UI Design

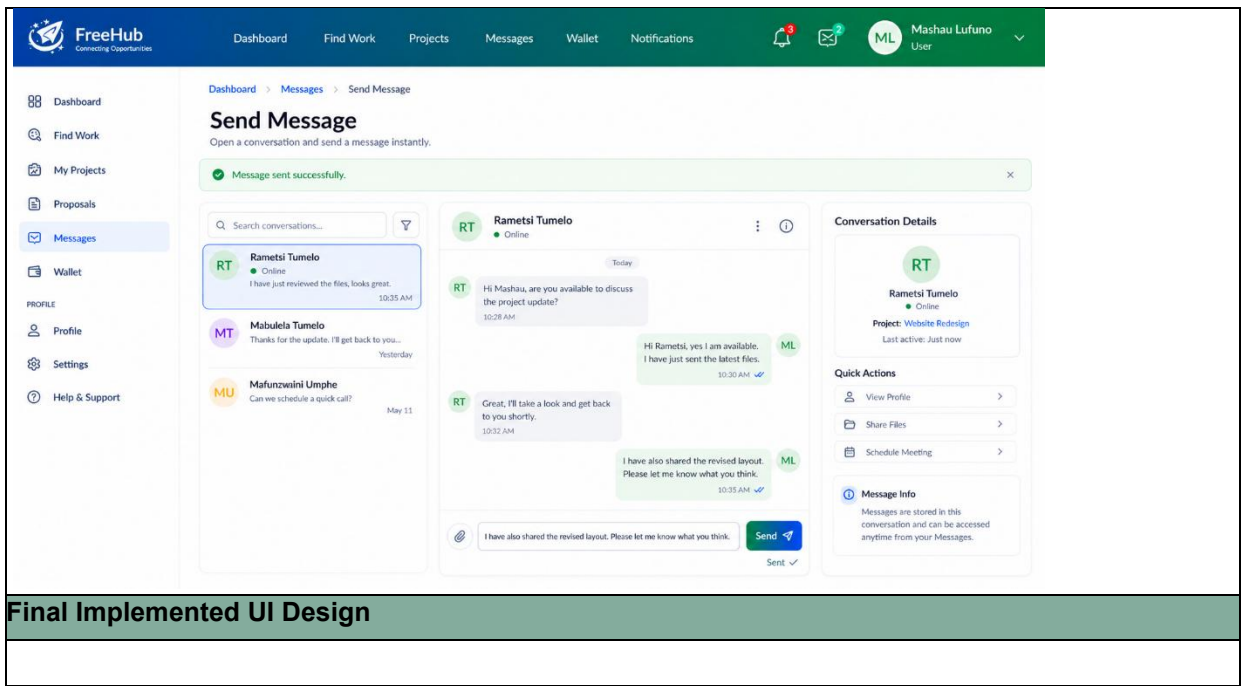
2.3.2.3 Designed by Lufuno Naithern Mashau

Use Case ID	Use Case Name
C100	Start Chat
Actors	
User	
Pre-Conditions	The user is logged in and has selected another user.
Triggers	The user selects "Start Chat".
Post-Conditions	A chat session is opened.
Basic Flow of Events / Main Success Scenario	
1. The user selects another user. 2. The user selects Start Chat. 3. The system creates or opens a chat session. 4. The system displays a chat interface.	
Initial UI Design	



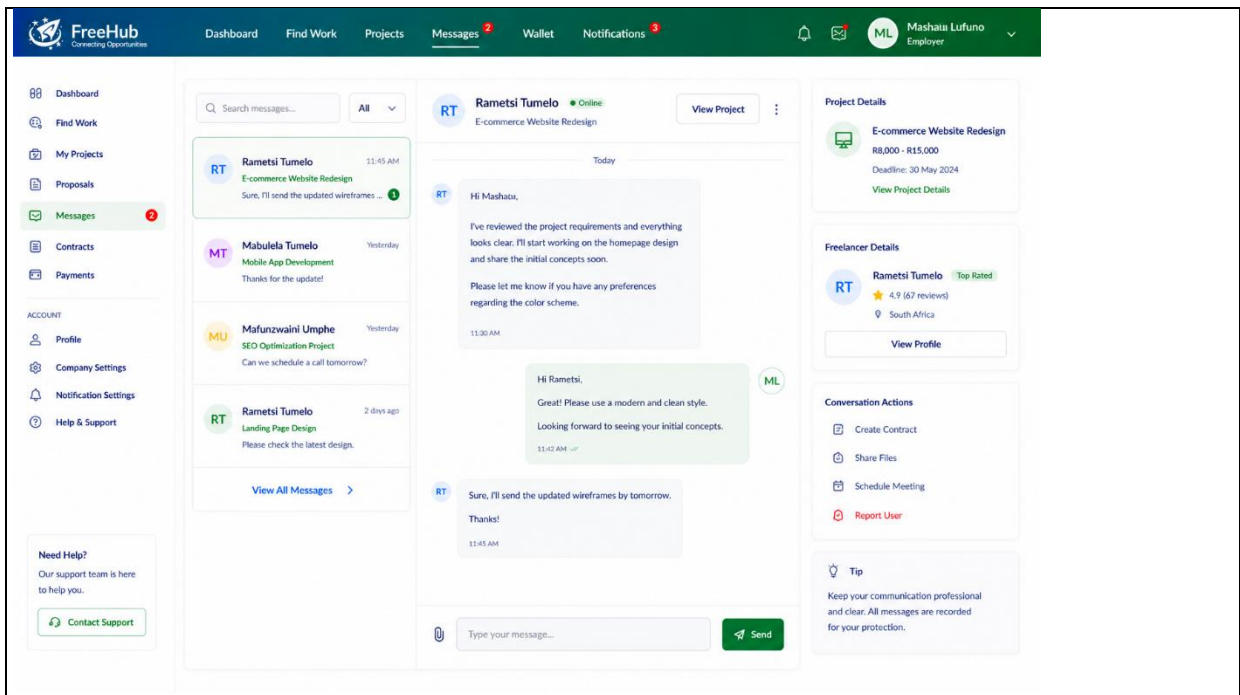
Final Implemented UI Design

Use Case ID	Use Case Name
C200	Send Message
Actors	
User	
Pre-Conditions	The user is logged in and has selected a conversation.
Triggers	The user types and sends a message.
Post-Conditions	The message is sent and stored.
Basic Flow of Events / Main Success Scenario	
1. The user opens a conversation. 2. The user enters a message. 3. The user selects Send. 4. The system validates the message. 5. The system sends messages to the receiver. 6. The system stores the message in the conversation.	
Initial UI Design	



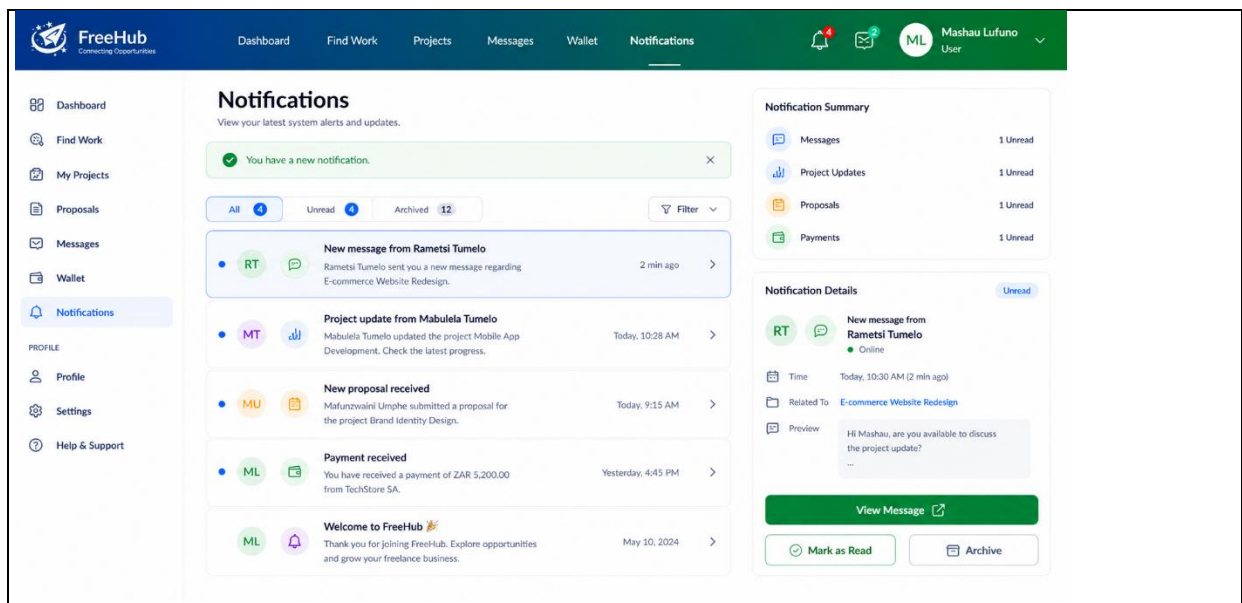
Final Implemented UI Design

Use Case ID	Use Case Name
C300	Receive Message
Actors	
User	
Pre-Conditions	The user has an active account.
Triggers	Another user sends a message.
Post-Conditions	The message appears in the user's conversation.
Basic Flow of Events / Main Success Scenario	
- Another user sends a message. - The system receives the message. - The system stores the message. - The system displays the message in the conversation.	
Initial UI Design	



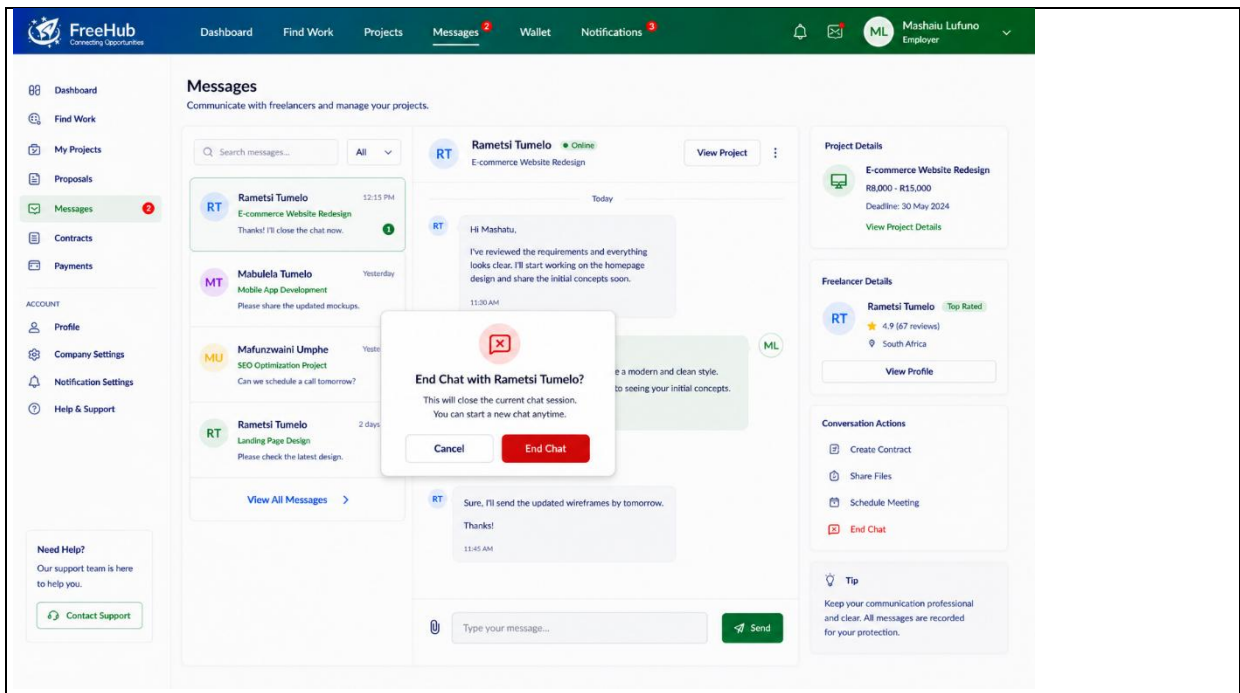
Final Implemented UI Design

Use Case ID	Use Case Name
C400	Receive Notification
Actors	
User	
Pre-Conditions	The user has an active account.
Triggers	A system event occurs, such as a new message or project update.
Post-Conditions	The notification is displayed to the user.
Basic Flow of Events / Main Success Scenario	
1. A new event occurs in the system. 2. The system creates a notification. 3. The system sends notifications to the user. 4. The user views the notification.	
Initial UI Design	



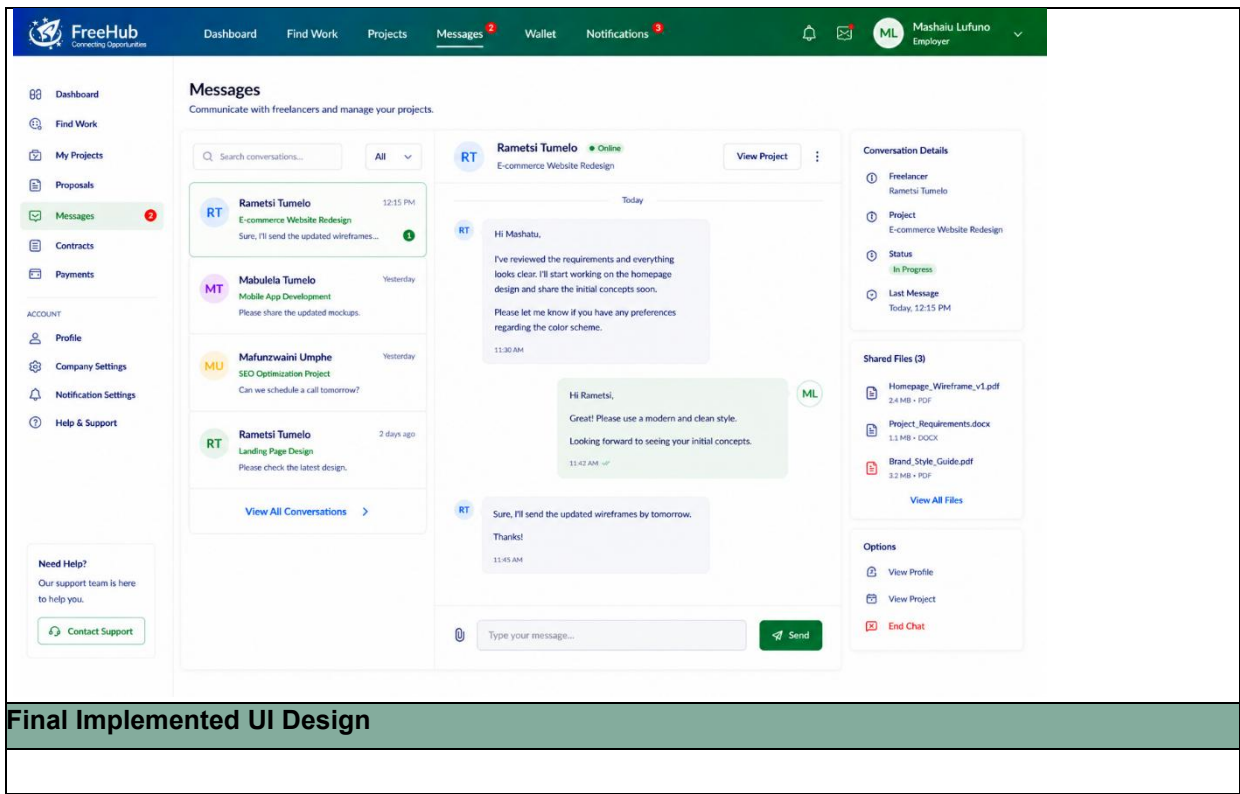
Final Implemented UI Design

Use Case ID	Use Case Name
C500	End Chat
Actors	
User	
Pre-Conditions	The user is in an active chat.
Triggers	The user selects "End Chat".
Post-Conditions	The chat session is closed.
Basic Flow of Events / Main Success Scenario	
1. The user opens an active chat. 2. The user selects End Chat. 3. The system closes the chat session. 4. The system returns the user to the conversation list.	
Initial UI Design	



Final Implemented UI Design

Use Case ID	Use Case Name
C600	View Conversation
Actors	
User	
Pre-Conditions	The user is logged in and has existing conversations.
Triggers	The user selects “View Conversation”.
Post-Conditions	The conversation history is displayed.
Basic Flow of Events / Main Success Scenario	
1. The user selects View Conversation. 2. The system retrieves conversation history. 3. The system displays a list of conversations. 4. The user selects a conversation. 5. The system displays full conversation.	
Initial UI Design	



Final Implemented UI Design

Use Case ID	Use Case Name
C700	Rate Employer
Actors	
Freelancer	
Pre-Conditions	The project is marked as completed.
Triggers	The freelancer selects the employer to rate from the completed projects list.
Post-Conditions	The employer's reputation score is automatically updated.
Basic Flow of Events / Main Success Scenario	
1. The freelancer selects a completed project from the completed projects list. 2. The system displays a rating form with fields for 1–5-star ratings and an optional comment. 3. The freelancer enters the rating and optional comment. 4. The system validates that all rating fields and complete. 5. The freelancer selects the Submit Rating option. 6. The system updates the employer's average rating and reputation score. 7. The system displays a pop-up message confirming successful rating submission. 8. The system returns the freelancer to the completed projects list.	
Initial UI Design	

FreeHub

Connecting Opportunities

Dashboard

Find Work

My Projects

Active Projects

Completed Projects

Cancelled Projects

Proposals

Messages

2

Wallet

ACCOUNT

Profile

Settings

Notification Settings

Help & Support

Need Help?

Our support team is here to help you.

Contact Support

My Projects

Completed Projects

E-commerce Website Redesign

Rate Employer

Your feedback helps build a trusted community. Please rate your experience working with this employer.

Project Completed

You marked this project as completed on 25 May 2024.

Project & Employer Summary

E-commerce Website Redesign

Completed on: 25 May 2024

Total Amount: R12,000

Employer

Mabulela Tumelo

South Africa

View Profile

Your Rating

Overall Rating *

Click a star to rate (1 = Poor, 5 = Excellent)

Comment (Optional)

Great communication and provided clear requirements.
Payment was on time. Would be happy to work with them again!

115/500 characters

Cancel

Submit Rating

Once submitted, your rating will be used to update the employer's reputation score.

About Ratings

Your honest feedback helps other freelancers make informed decisions.

Employers with higher ratings tend to get better quality freelancers.

All ratings are private and reviewed for quality.

Rating Guidelines

Be honest and fair

Focus on your experience

Avoid personal information

Keep comments professional

Need Help?

If you have an issue with this employer, you can report them.

Report Employer

Final Implemented UI Design

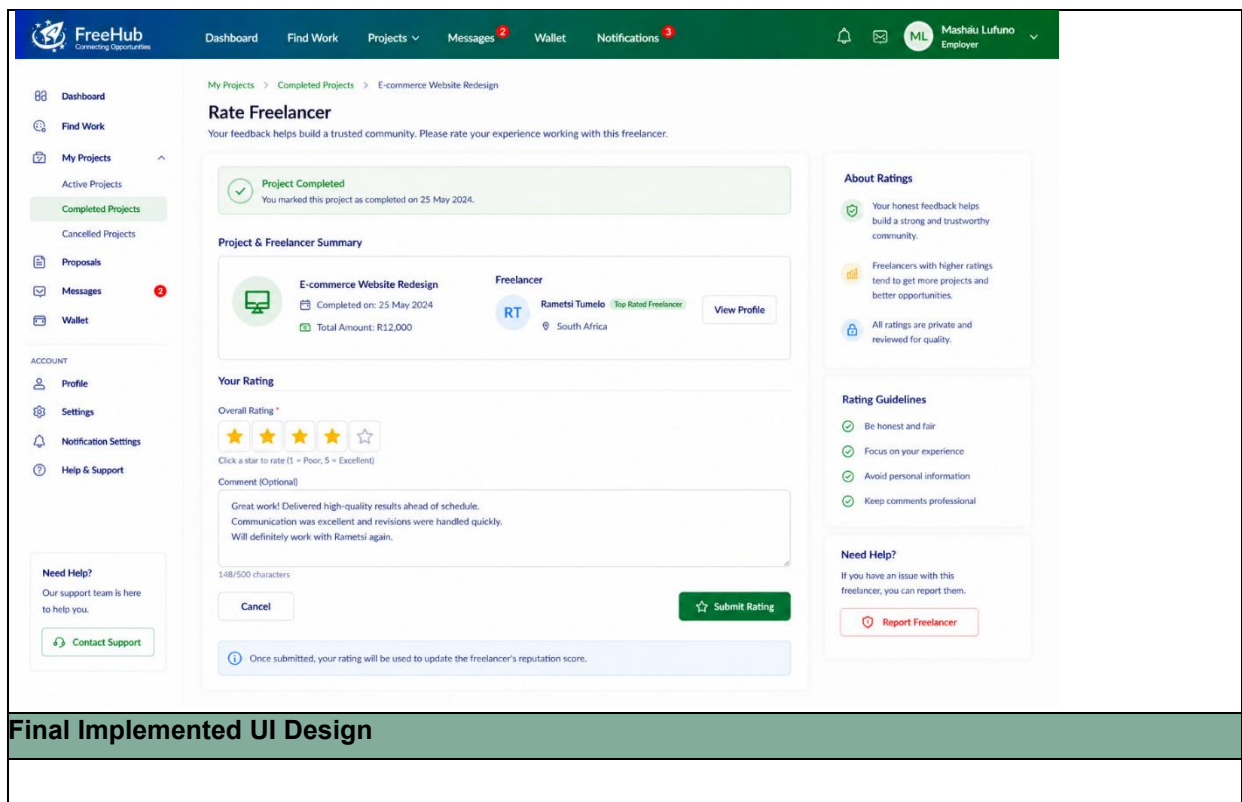
Use Case ID	Use Case Name
C800	Rate Freelancer
Actors	
Employer	
Pre-Conditions	The project is marked as completed.
Triggers	The employer selects the freelancer to rate from the completed projects list.
Post-Conditions	The freelancer's reputation score is automatically updated.

Basic Flow of Events / Main Success Scenario

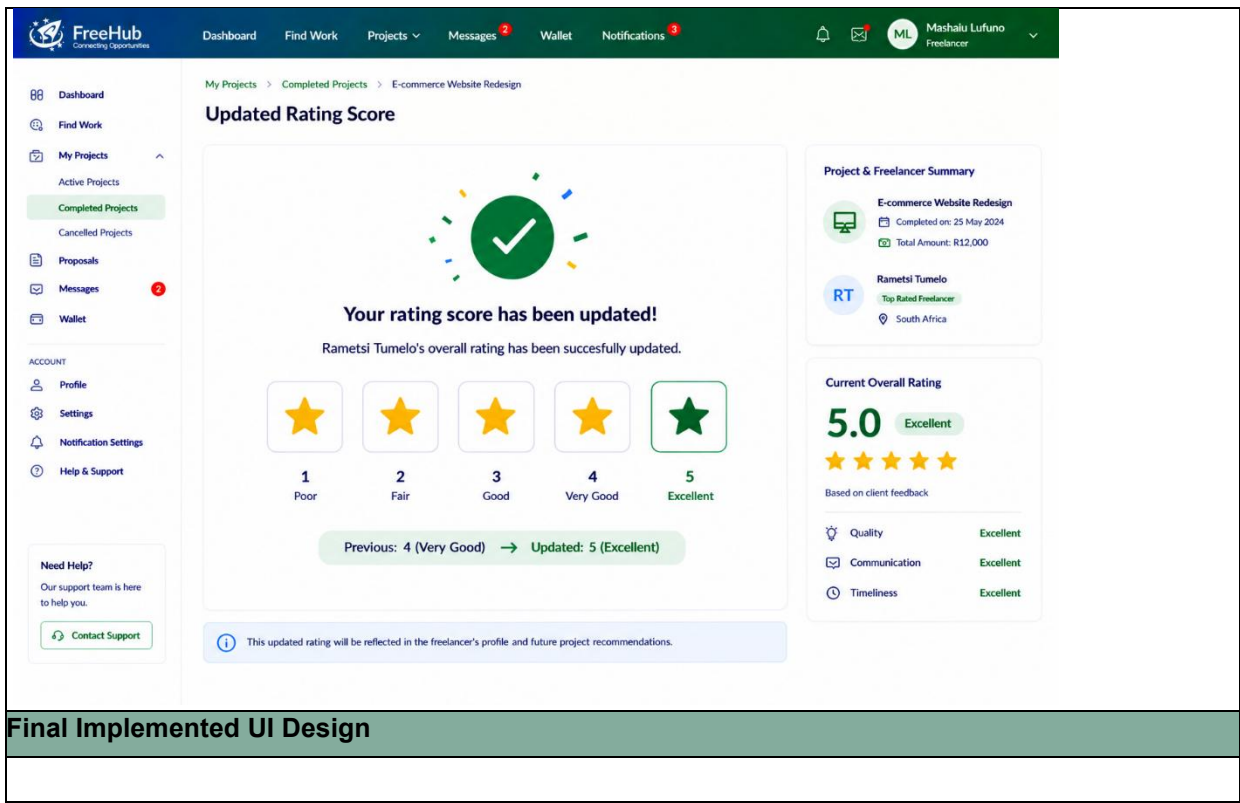
- The employer selects a completed project from the completed projects list.
- The system displays a rating form with fields for 1-5 ratings and an optional comment field.
- The employer enters the rating and optional comment.
- The system validates that all ratings are complete.
- The employer selects the "Submit Rating" option.
- The system updates the freelancer's average rating and reputation score.
- The system displays a message confirming successful rating submission.
- The system returns the employer to the completed projects list.

Initial UI Design

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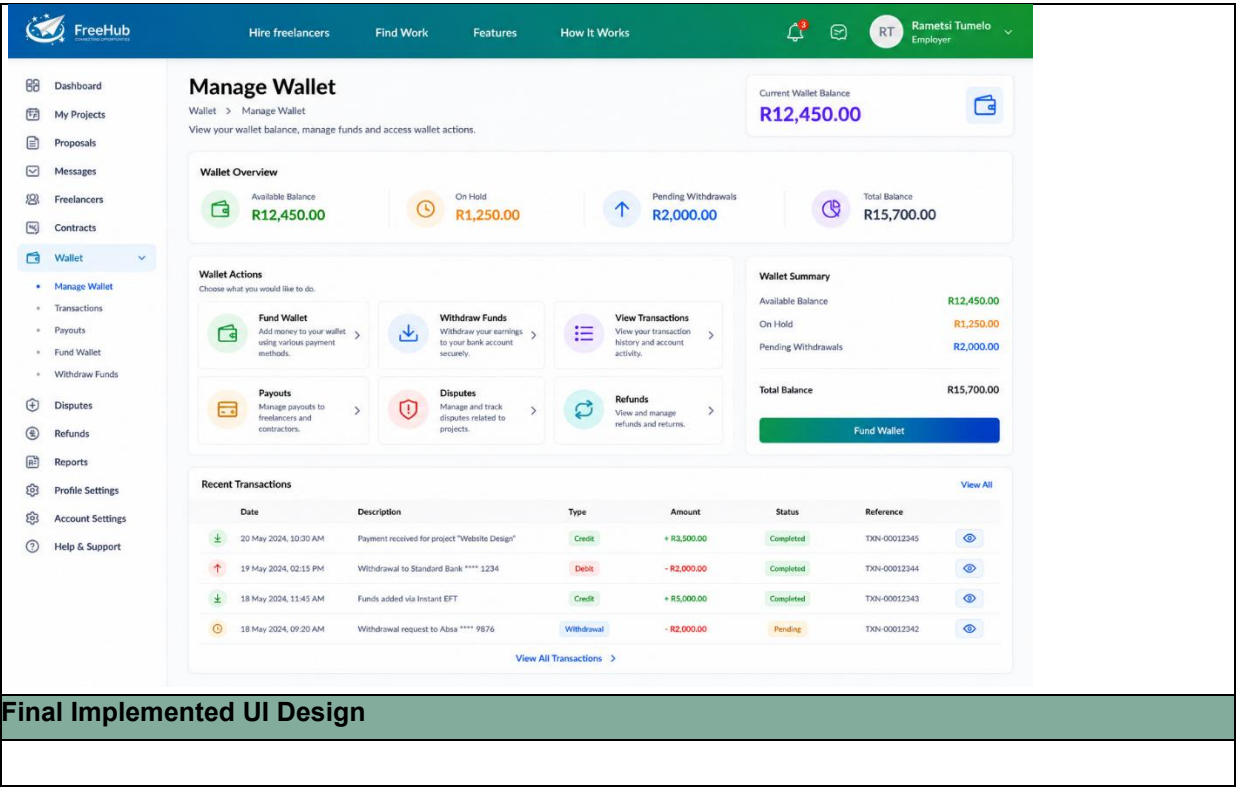


Use Case ID	Use Case Name
C900	Update Rating Score
Actors	
System	
Pre-Conditions	A rating has been submitted.
Triggers	The system receives a new rating.
Post-Conditions	The user's rating score is updated.
Basic Flow of Events / Main Success Scenario	
1. The system receives a submitted rating. 2. The system validates the rating. 3. The system calculates the updated average rating. 4. The system updates the user's rating score. 5. The system displays the updated rating.	
Initial UI Design	



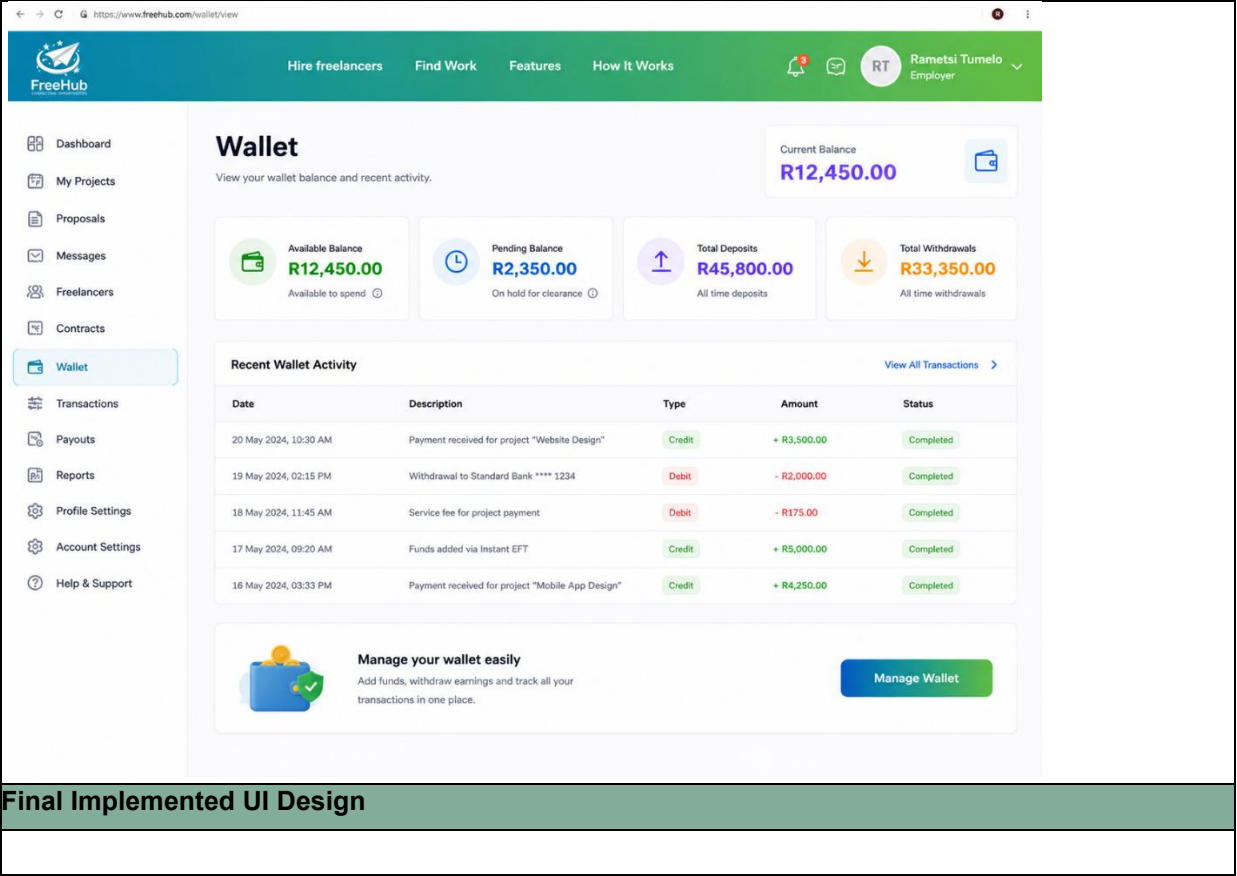
2.3.2.4 Designed by Tumelo Gift Rametsi

Use Case ID	Use Case Name
D100	Manage Wallet
Actors	
User	
Pre-Conditions	The user is logged in and has a wallet.
Triggers	The user selects “Manage Wallet”.
Post-Conditions	Wallet settings are updated.
Basic Flow of Events / Main Success Scenario	
1. The user selects Manage Wallet. 2. The system displays wallet options. 3. The user chooses an action such as view balance, fund wallet, withdraw funds, or view transactions. 4. The system performs the selected action. 5. The system displays the updated wallet information.	
Initial UI Design	



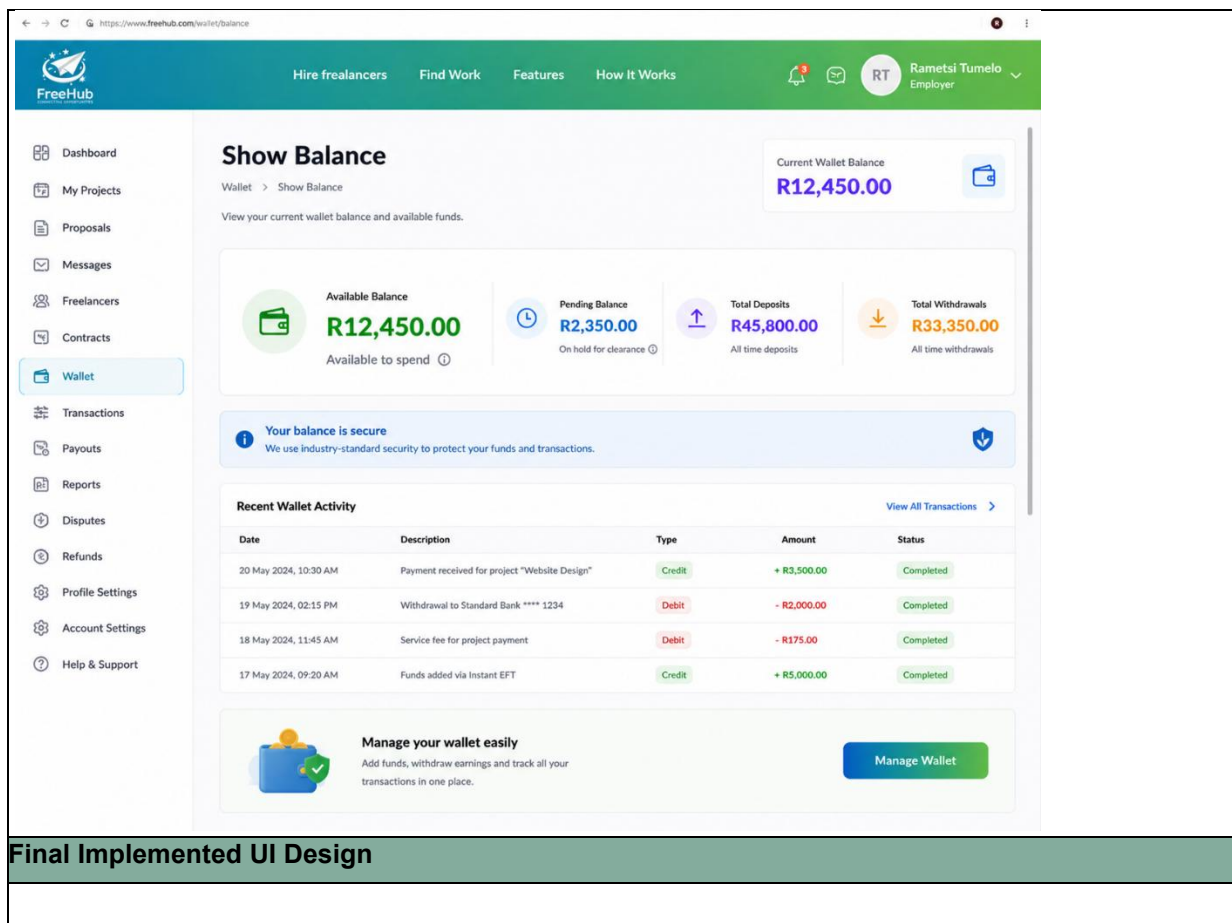
Final Implemented UI Design

Use Case ID	Use Case Name
D200	View Wallet
Actors	
User	
Pre-Conditions	The user is logged into the system.
Triggers	The user selects “Wallet”.
Post-Conditions	The wallet details are displayed.
Basic Flow of Events / Main Success Scenario	
1. The user selects the Wallet option. 2. The system retrieves the user’s wallet information. 3. The system displays the wallet balance and recent wallet activity.	
Initial UI Design	



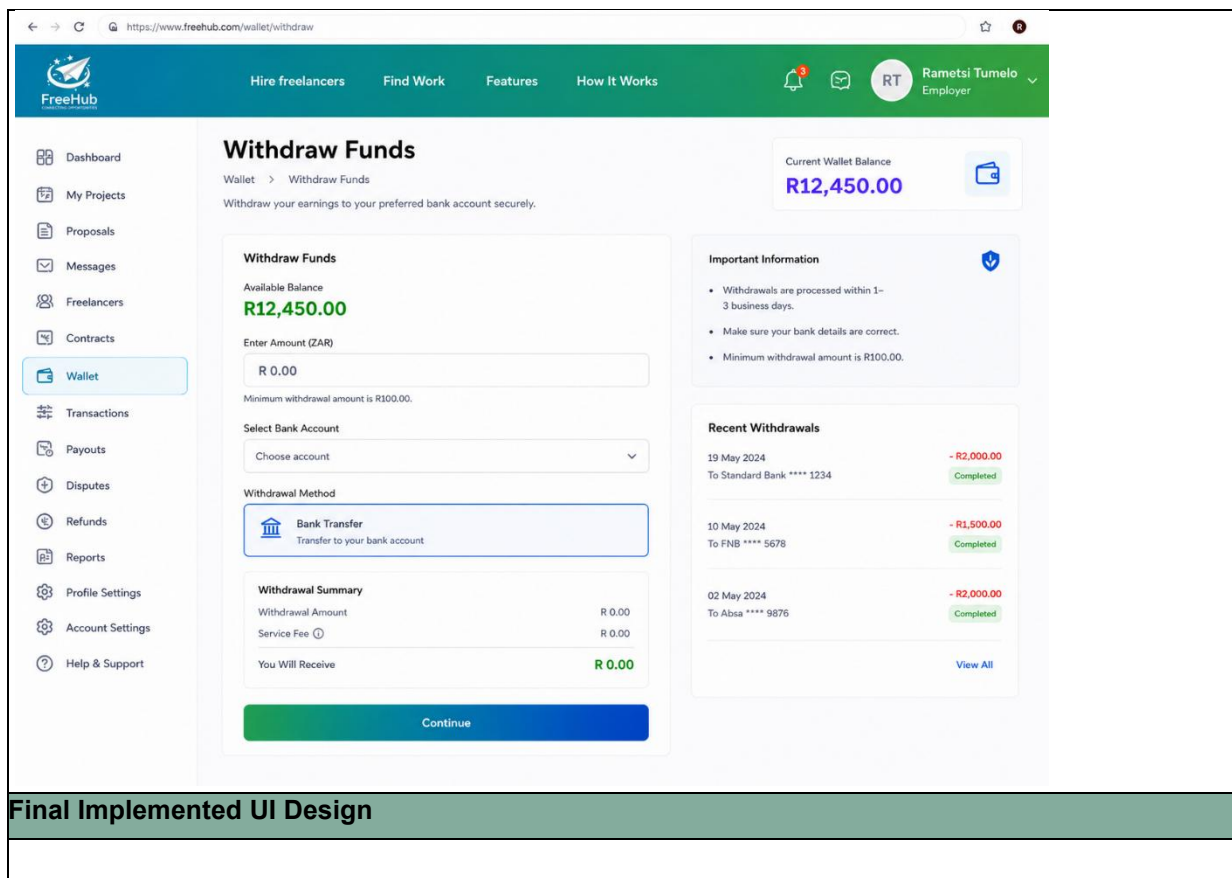
Final Implemented UI Design

Use Case ID	Use Case Name
D300	Show Balance
Actors	
User	
Pre-Conditions	The user is logged in.
Triggers	The user selects “Show Balance”.
Post-Conditions	The current wallet balance is shown.
Basic Flow of Events / Main Success Scenario	
1. The user selects Show Balance.	
2. The system retrieves the current wallet balance.	
3. The system displays the available balance.	



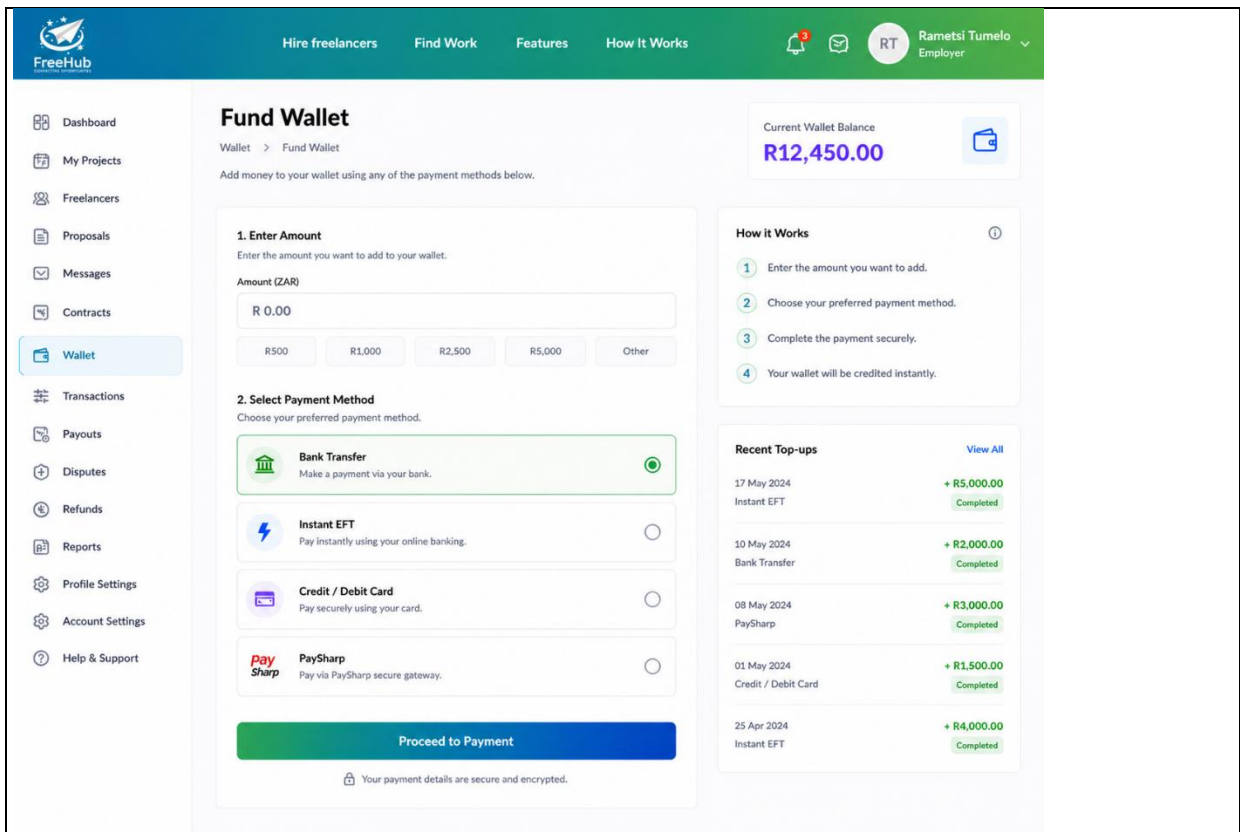
Final Implemented UI Design

Use Case ID	Use Case Name
D400	Withdraw Funds
Actors	
User	
Pre-Conditions	The user is logged in and has available funds.
Triggers	The user selects "Withdraw Funds".
Post-Conditions	A withdrawal request is created.
Basic Flow of Events / Main Success Scenario	
- The user selects Withdraw Funds. - The system displays a withdrawal form. - The user enters the amount and banking details. - The system validates the amount and details. - The user confirms the withdrawal. - The system processes the withdrawal request. - The system displays a confirmation message.	
Initial UI Design	



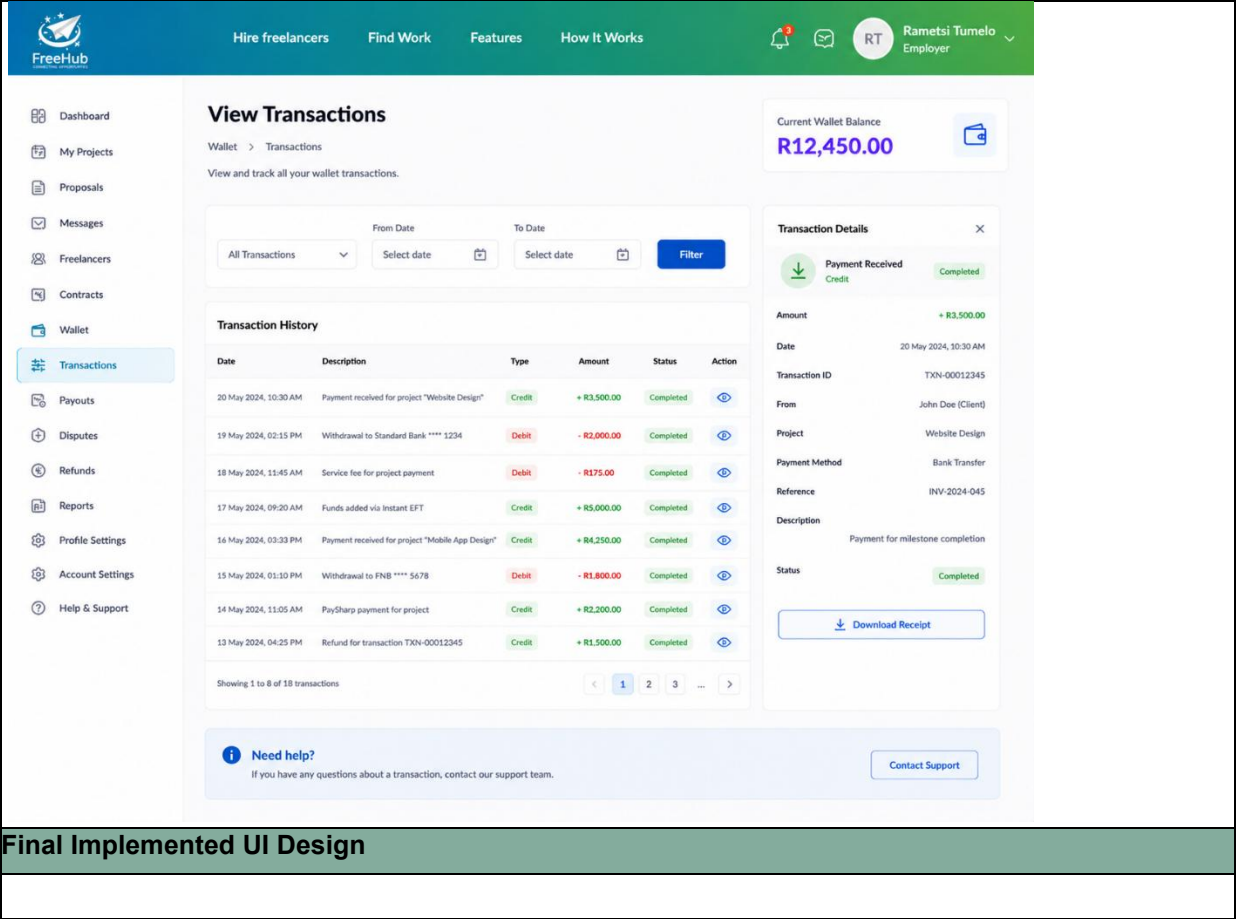
Final Implemented UI Design

Use Case ID	Use Case Name
D500	Fund Wallet
Actors	
Employer	
Pre-Conditions	The Employer is logged in
Triggers	The Employer selects "Fund Wallet".
Post-Conditions	The wallet balance is updated.
Basic Flow of Events / Main Success Scenario	
1. The Employer selects Fund Wallet. 2. The system displays payment options. 3. The Employer enters the amount to deposit. 4. The Employer selects a payment method. 5. The system processes the payment. 6. The system updates the wallet balance. 7. The system displays a confirmation message.	
Initial UI Design	

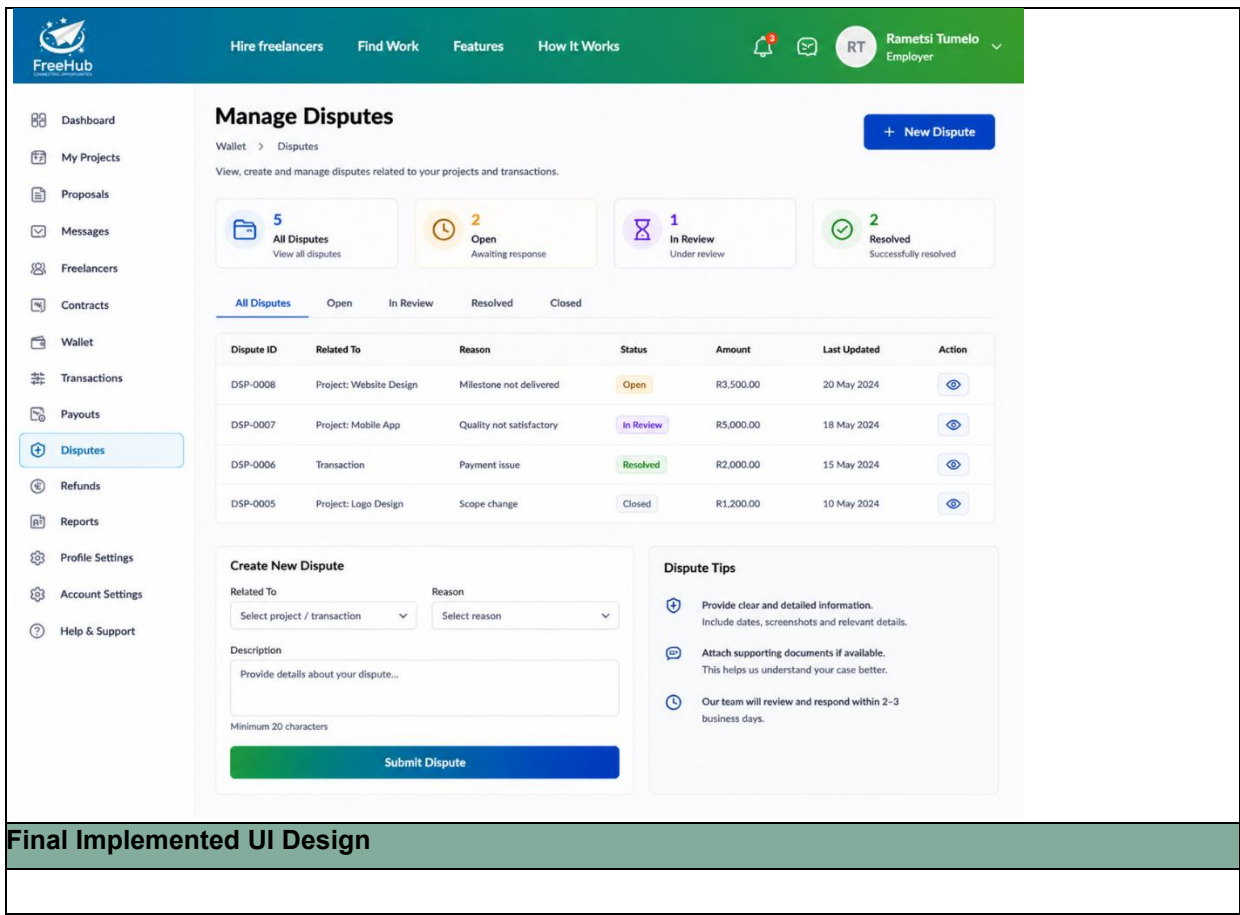


Final Implemented UI Design

Use Case ID	Use Case Name
D600	View Transactions
Actors	
User	
Pre-Conditions	The user is logged in.
Triggers	The user selects “View Transactions”.
Post-Conditions	Transaction history is displayed.
Basic Flow of Events / Main Success Scenario	
1. The user selects View Transactions. 2. The system retrieves transaction history. 3. The system displays a list of transactions. 4. The user selects a transaction to view more details. 5. The system displays the selected transaction details.	
Initial UI Design	



Use Case ID	Use Case Name
D700	Manage Disputes
Actors	
User	
Pre-Conditions	The user is logged in and has an existing transaction or project.
Triggers	The user selects "Manage Disputes".
Post-Conditions	A dispute is created, updated, or resolved.
Basic Flow of Events / Main Success Scenario	
1. The user selects Manage Disputes. 2. The system displays existing disputes. 3. The user selects a dispute or creates a new one. 4. The user enters dispute details. 5. The system saves the dispute information. 6. The system displays a confirmation message.	
Initial UI Design	



Final Implemented UI Design

Use Case ID	Use Case Name
D800	Trigger Refunds
Actors	
Employer	
Pre-Conditions	The Employer is logged in and has a valid refundable transaction
Triggers	The employer selects “Refund”.
Post-Conditions	A refund request is submitted.
Basic Flow of Events / Main Success Scenario	
1. The user selects a transaction. 2. The user selects Trigger Refund. 3. The system displays refund details. 4. The user confirms the refund request. 5. The system validates the refund eligibility. 6. The system submits the refund request. 7. The system displays a confirmation message	
Initial UI Design	

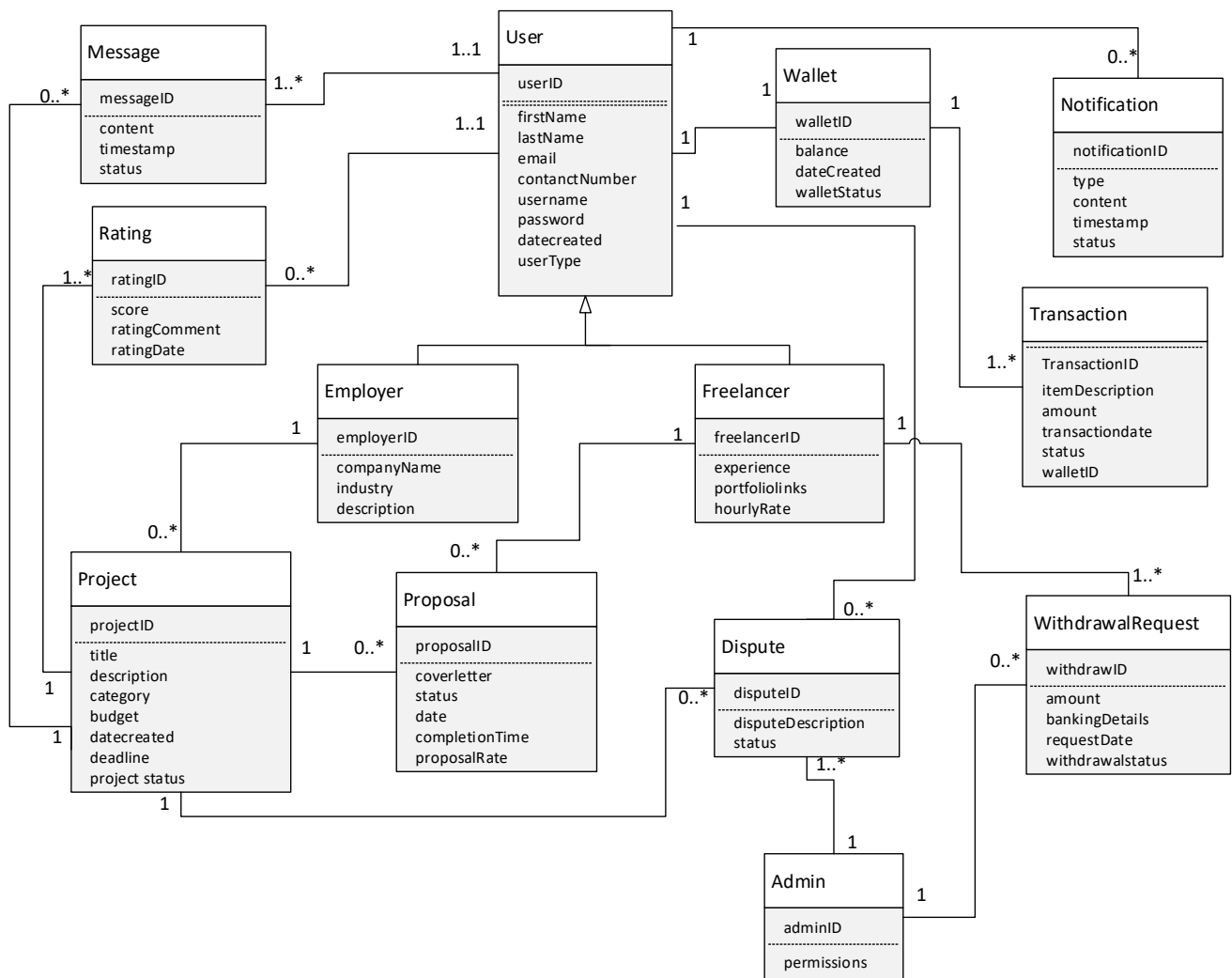
3 DATA REQUIREMENTS

3.1 List of data and attributes

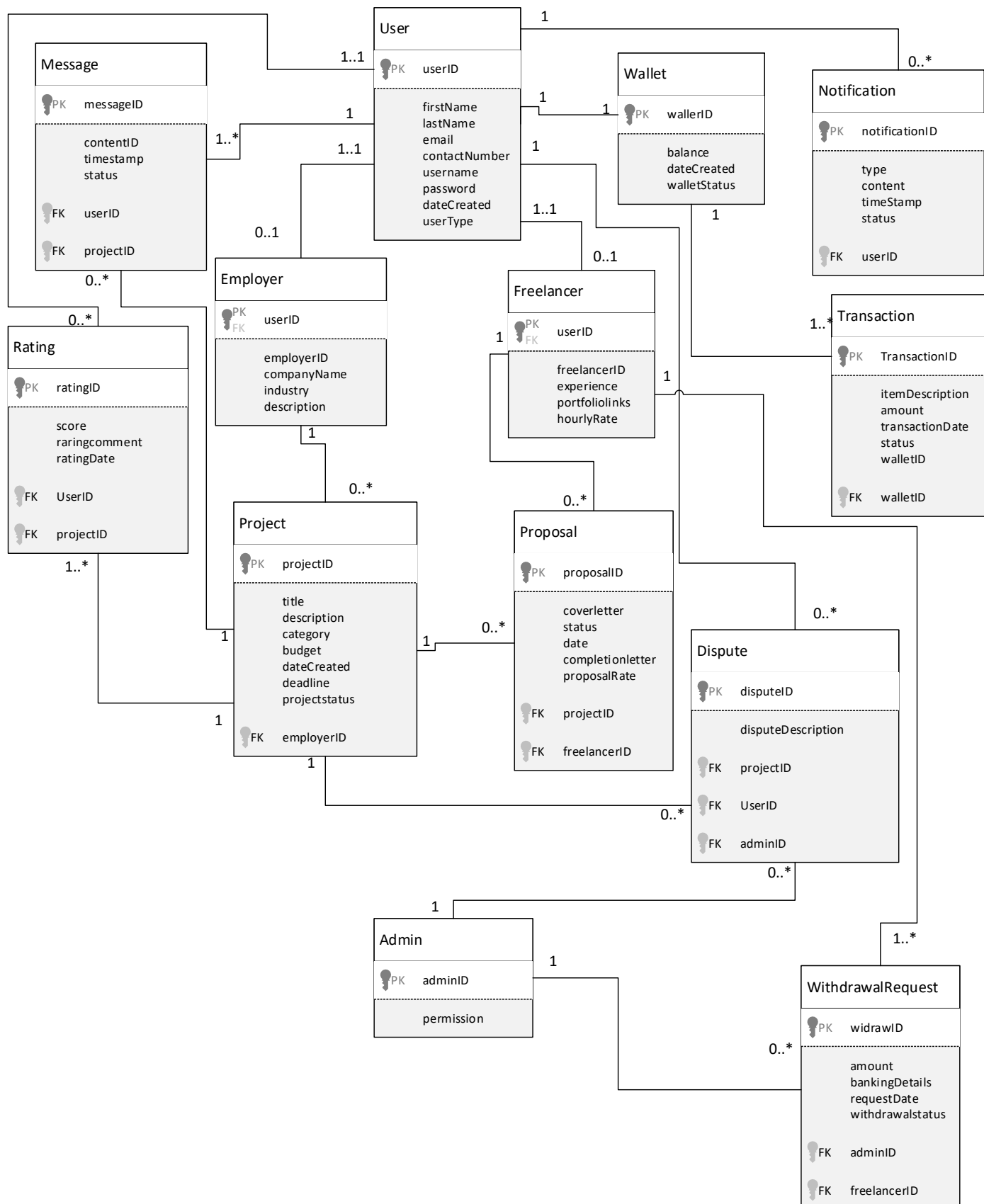
Entity	Entity Attributes
User	userID firstName lastName email contactNumber username password accountStatus dateCreated ratingScore
Employer	userID employerID companyName industry description
Freelancer	freelancerID userID skills experience portfoliolinks hourlyRate
Wallet	walletID userID balance dateCreated walletSatus
Transaction	transactionID walletID itemDescription transactionType transactionAmount transactionDate transactionStatus
Project	projectID projectTitle employerID title description category budget dateCreated deadline projectStatus

Proposal	proposalID projectID freelancerID coverLetter status date estimatedCompletionTime proposedRate
Rating	ratingID projectID score ratingComment ratingDate
Dispute	disputeID userID projectID transactionID disputeDescription status
Message	messageID senderID recieverID projectID content timeStamp status
Notification	NotificationID UserID Type Content Timestamp Status
Admin	adminID permissions
ChatSession	sessionID participants ProjectID startTime EndTime status
Comment	commentID commentText commentDate
Conversation	conversationID startDate endDate conversationStatus
WithdrawalRequest	withdrawID amount bankingDetails requestDate withdrawalStatus

3.2 Domain Class Diagram



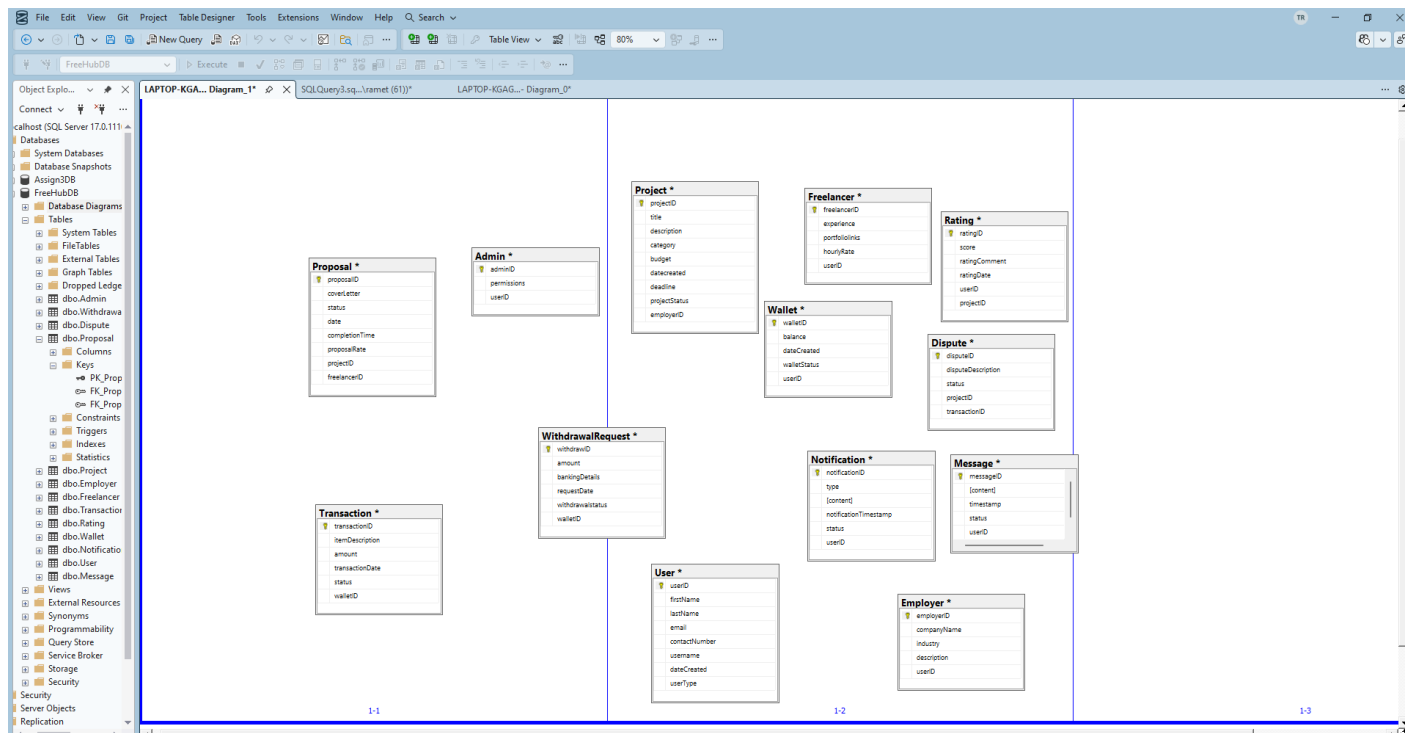
3.3 Draft Implementation Ready Class Diagram



3.4 Implemented database structure

We have used **SQL Server** to create our database in.

The name of the database for our project is **FreeHubDB**



SQL Query 3.s...vramet (61)*

```
SELECT
1  fk.name AS ForeignKeyName,
2  OBJECT_NAME(fk.parent_object_id) AS ChildTable,
3  COL_NAME(fkc.parent_object_id, fkc.parent_column_id) AS ChildColumn,
4  OBJECT_NAME(fk.referenced_object_id) AS ParentTable,
5  COL_NAME(fkc.referenced_object_id, fkc.referenced_column_id) AS ParentColumn
6  FROM sys.foreign_keys fk
7  INNER JOIN sys.foreign_key_columns fkc
8  ON fk.object_id = fkc.constraint_object_id
9  ORDER BY ChildTable, ForeignKeyName;
```

100 % No issues found

ForeignKeyName	ChildTable	ChildColumn	ParentTable	ParentColumn
FK_Admin_User	Admin	userID	User	userID
FK_Dispute_Project	Dispute	projectID	Project	projectID
FK_Dispute_Transaction	Dispute	transactionID	Transaction	transactionID
FK_Employer_User	Employer	userID	User	userID
FK_Freelancer_User	Freelancer	userID	User	userID
FK_Message_User	Message	userID	User	userID
FK_Notification_User	Notification	userID	User	userID
FK_Project_Employer	Project	employerID	Employer	employerID
FK_Proposal_Freelancer	Proposal	freelancerID	Freelancer	freelancerID
FK_Proposal_Project	Proposal	projectID	Project	projectID
FK_Rating_Project	Rating	projectID	Project	projectID
FK_Rating_User	Rating	userID	User	userID
FK_Transaction_Wallet	Transacti...	walletID	Wallet	walletID
FK_Wallet_User	Wallet	userID	User	userID
FK-WithdrawalReques...	Withdraw...	walletID	Wallet	walletID

Query executed successfully.

4 TECHNICAL REQUIREMENTS

4.1 System Environment

The Free HUB system is a website that does not need internet connection to work and it can be used offline. The backend is built using C# (.NET Core), which handles the functionality of the system like logic business and user management. Java will also be used for other background task like the calculation of reputation scores and also handling disputes.

The frontend will be created using HTML, CSS and JavaScript to make the website more simple, interactive, fast and compatible with most browsers. All Information such as user profiles, projects, proposals, messages, ratings and wallets transactions will be stored in the Microsoft SQL server as the database.

The system will also be hosted on the internet information Services (IIS) and uses HTTPS encryption to keep user data safe.

4.2 Start-up Instructions

To run the FreeHUB project specifically using Visual Studio 2022 or latest, follow these technical steps to ensure the environment is configured correctly for your .NET Core and SQL Server implementation.

1. Prerequisites

- **Visual Studio 2022 or later:** Ensure the ".NET desktop development" and "ASP.NET and web development" workloads are installed.
- **SQL Server:** The database must be attached to Microsoft SQL Server as defined in your Data Requirements.

2. Opening and Configuring the Project

- Open the Solution:
- Navigate to your project folder and locate the file named FreeHUB.sln.
- Double-click it to open the entire project structure in Visual Studio.

3. Database Connection:

- Open the appsettings.json file.
- Update the ConnectionStrings to match your local SQL Server instance name.

4. Running the System

- Press F5 (Debug mode) or Ctrl+F5 (Run without debugging).

Visual Studio will build the C# logic and launch your default browser to the FreeHUB home page

4.3 Logins & Passwords

Login details for all users are:

- **Admin:**

Username: admin@freehub.co.za

Password: admin@2026

- **Freelancer:**

Username: freelancer@freehub.co.za

Password: free@2026

- **Employer:**

Username: employer@freehub.co.za

Password: emp@2026